

Techno-economic feasibility of a PV/battery/fuel cell/electrolyzer/biogas hybrid system for energy and hydrogen production in the far north region of cameroon by using HOMER pro

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ABSTRACT

The electricity deficit in Cameroon is estimated today at 50 GWh. This deficit characterized by frequent and sometimes prolonged load shedding, disrupts economic and social life. To overcome this electricity deficit, Cameroon took the decision to produce 3000 MW of electrical energy from its renewable energies potential. Indeed, the annual solar radiation in Cameroon varies from 4.28 kWh/m²/year to 5.80 kWh/m²/year. It has 25 million hectares of forest covering three-quarters of its territory, amounting to the third-largest biomass potential in sub-Saharan Africa. In addition, there is an intense breeding activity for cattle, goats, sheep and pigs in the Far North region amounting to several million heads and generating large quantities of dung. This paper therefore investigates for the very first time the techno-economic feasibility by using HOMER Pro of two scenarios of hybrid systems namely, PV/Fuel Cell/Electrolyzer/Biogas (scenario 1), and PV/Battery/Fuel Cell/Electrolyzer/Biogas (scenario 2) for energy and Hydrogen production in the city of Maroua, recognized as being part of the sunniest region (Far North) of Cameroon. The combination of electrolyzer, fuel cell and hydrogen tank was used in the present design to reduce battery storage requirement. Three types of household electricity demands communities (low, medium and high consumers) were considered in this work. The results showed that the optimal system architecture for scenario 1 included 144 kW PV module, 15 kW biogas generator, 11 kW converter, 15 kW electrolyzer, 15 kW fuel cell and 5000 kg Hydrogen tank with a dispatch strategy of cycle charging (CC) for low consumers community. For medium consumers community of scenario 1, 879 kW PV module, 15 kW biogas generator, 31.9 kW converter, 24 kW fuel cell, 24 kW electrolyzer and 5000 kg Hydrogen tank with a CC dispatch strategy was the best hybrid system. For high consumers community of scenario 1, 11,925 kW PV module, 15 kW biogas generator, 570 kW converter, 266 kW fuel cell, 266 kW electrolyzer and 25,000 kg Hydrogen tank with a CC dispatch strategy was the best hybrid system. Concerning scenario 2, the following architectures were the best hybrid systems: for low consumers, 138 kW PV modules, 15 kW biogas generator, 27.2 kW converter, 15 kW fuel cell, 15 kW electrolyzer, 5000 kg Hydrogen tank and 480 batteries storage bank with a CC dispatch strategy; for medium consumers, 234 kW PV modules, 15 kW biogas generator, 57.8 kW converter, 24 kW fuel cell, 24 kW electrolyzer, 5000 kg Hydrogen tank and 1023 batteries storage bank with a load following (LF) dispatch strategy; and for high consumers, 820 kW PV modules, 15 kW biogas generator, 405 kW converter, 266 kW fuel cell, 266 kW electrolyzer, 25,000 kg Hydrogen tank and 9519 batteries storage bank with a CC dispatch strategy. The levelized costs of energy (LCOE) for scenario 1 were US\$ 0.871/kWh, US\$ 0.898/kWh and US\$ 1.524/kWh for low, medium and high consumers communities respectively. Concerning scenario 2, the LCOE were US\$ 0.139/kWh, US\$ 0.091/kWh and US\$ 0.071/kWh. In addition, it was found a levelized costs of Hydrogen (LCOH) for scenario 1 of US\$ 7.66/kg, US\$ 4.95/kg, and US\$ 0.45/kg for low, medium and high consumers communities respectively. For scenario 2, the LCOH were US\$ 3.06/kg, US\$ 1.34/kg and US\$ 0.15/kg for low, medium and high consumers communities respectively. It was also concluded from the optimization results that the combination of water electrolyzer, fuel cell and hydrogen tank

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coupled to biogas generator and PV modules could be used as an alternative solution to make electricity available and accessible to the population of the Far North region of Cameroon.

1. Introduction

While energy is the major pillar of all development, Cameroon, a central Africa country [1] does not always take advantage of the cutting-edge technologies available to science to solve this unavoidable development concern. Electricity remains a luxury product for thousands of Cameroonians and in particular for disadvantaged populations [2]. The issue of renewable energies and their hybridization, which is increasingly present in the public debate [3], is a great opportunity to be seized for the electrification of Cameroonian cities. Indeed, a plethora of research works emerged in the domain of hybrid energy systems. Rahman and Tam studied the feasibility of applying fuel cell (FC) to provide operational support to photovoltaic (PV) modules. The authors demonstrated here that the combination of FC and PV could be used to meet variable loads for either utility or stand-alone applications. It was also shown that the PV-FC association overcame the intermittency problem inherent with PV [4]. A two-loop controller was used to maximize the performance of a grid-connected PV/FC hybrid system [5]. The computations performed in their analysis revealed that the proposed approach allowed the PV array to accurately track the peak power conditions even at times of rapidly changing insolation. Das et al. proposed a hybrid energy system consisting of wind, PV and FC to supply continuous power to the load [6]. They showed that the proposed hybrid system allowed maximum utilization of freely available renewable energy sources like wind and PV and demand-based utilization of hydrogen-based FC. Their results presented the fact that the FC controller responded efficiently to the deficit energy demands. It was also demonstrated that with both wind and PV operating at their rated capacity, the system could produce power as high as 35 kW. An economic evaluation of a hybrid Wind/PV/FC generation system for a typical home in the Pacific Northwest was performed by Nelson et al. [7]. They noticed a clear economic advantage of PV/battery/wind system over the PV/wind/FC/electrolyzer system which indicated a need for extensive research in the FC/electrolyzer area. They also mentioned that by improving the efficiency of both FC and electrolyzer, it could be possible that a FC/electrolyzer storage system might be economically competitive. El-Shatter et al. developed an energy system comprising three energy sources, namely PV, wind and FC [8]. Their study demonstrated the feasibility of the use of fuzzy logic controllers to: track the maximum power available in both wind and PV energy systems; and regulate the DC bus voltage in the whole range of wind speeds and solar insolation. Alam and Gao proposed a hybrid energy generation system consisting of PV, wind turbine and FC adequate for remote areas [9]. The authors showed that the annual estimations of solar, wind and FC powers were 29.9%, 70.15% and 0.56% respectively. Onar et al. presented a hybrid system consisting of PV, FC, Wind and ultra-capacitor for grid-independent applications [10]. The novelty of their work was the hybridization of the renewable energies with FC using long and short term storage strategies with appropriate power controllers and control strategy. They concluded that the proposed system and its control strategy exhibited excellent performance for a particular period of time. Ahmed et al. proposed a combining variable speed wind turbine, solar PV and FC generation systems to supply continuous power to a typical 2 kW/150 V dc load as telecommunication power plants or ac residential power applications in isolated islands continuously throughout the year [11]. The solutions of their simulations indicated that even when the sun and wind were not present; the system was reliable. Zervas et al. indicated that the combination of PV, electrolyzer, metal hydride tanks, and proton exchange membrane FC was more advantageous compared to stand-alone PV systems because it was possible to store efficiently solar energy by transforming it into Hydrogen [12]. Hwang et al. modeled

mathematically a PV Hydrogen FC hybrid system to maximize the use of renewable energy sources [13]. Their system was made of PV, a water electrolyzer, a Hydrogen tank, and a proton exchange membrane FC generator. The FC generator was used as a backup generator to supplement the load demands when the PV power was deficient during a period of low solar radiation. The investigators deduced that the system favorably tracked the daily energy consumption of a typical residential building. In order to optimize the performance of a PV/FC/Battery/supercapacitor, Gebre et al. presented an optimal load sharing strategy [14]. The FC was utilized in their study to add the intermittent output of the PV while the battery storage was used to compensate for part of the temporary peak demand which the PV and FC could not meet thus avoiding oversizing of FC. Leva and Zaninelli introduced a PV/wind/diesel hybrid system to supply a remote power load [15]. The results illustrated that the power balance was assured and the system could operate properly with FC too. The PV-wind-FC allowed an increase of the sustainability of the project by considering the environmental aspect. Silva et al. performed an economic assessment and optimization of a PV/FC/Batteries hybrid system for use in the Brazilian Amazon [16]. It was shown that the PV-FC-battery configuration that utilized energy stored as Hydrogen proved to be technically feasible under the adverse climatic conditions of the Amazon. Ganguly et al. modeled and analyzed a greenhouse-integrated power system consisting of PV panels, electrolyzer bank and Polymer Electrolyte Membrane FC stacks [17]. Their results showed that 51 PV modules each of 75 W_p along with a 3.3 kW electrolyzer and 2 Polymer Electrolyte Membrane FC stacks, each of 480 W, could support the energy requirement of a 90 m² floriculture greenhouse with fan-pad ventilated system. A mobile renewable house using PV/wind/FC hybrid power was developed by Eroglu et al. [18]. They considered a combination of a PV module (800 W), a wind turbine (1 kW) and a FC (2 kW) to produce a maximum power of 3.8 kW. The excess of energy produced was converted to Hydrogen via an electrolyzer and stored in a high pressure Hydrogen tank. Karakoulidis et al. modeled using HOMER software a stand-alone PV-diesel-battery-FC to meet the electric load of a machinery laboratory located in Kavala town in Greece [19]. The simulated solutions indicated that the replacement of diesel generators and batteries by Hydrogen was technologically feasible, reduced emissions, noise and fossil fuel dependence and increased renewable energies penetration. Thounthong et al. developed a control algorithm for a FC/solar cell/supercapacitor hybrid power source [20]. They demonstrated that a polymer electrolyte membrane FC (1200 W, 46 A) manufactured by the Ballard Power System Company, a PV array (800 W, 31 A) manufactured by the Ekarat Solar Company and a supercapacitor module (100 F, 32 V) manufactured by the Maxwell Technologies Company, illustrated the excellent energy-management scheme during load cycles. Silva et al. examined the use of a PV/FC/battery hybrid system to supply the electricity in an isolated community in Brazil [21]. Their analysis showed that the optimal system's initial cost, net present cost, and electricity cost with the Hydrogen storage system were US\$ 87,138; US\$102,323; and US\$ 1.351/kWh, respectively. Rekioua et al. developed a hybrid PV/FC hybrid system for stand-alone application [22]. Their system consisted of a PV module (primary source), which converted solar irradiation into electrical energy; the electrolyzer which generated Hydrogen and oxygen from water and a FC which utilized the generated Hydrogen to produce electricity. Mohammed et al. proposed an optimal design of a PV/FC hybrid system without battery for the city of Brest in France [23]. It was concluded that their proposed hybrid power system was a viable solution (non-polluting reliable energy source) compared to diesel generators. It was also shown that a FC could efficiently complement a fluctuating renewable source as solar energy to satisfy growing loads.

Mokrani et al. presented a modeling, control and power management of PV/FC/Battery to supply electric vehicle in Algeria [24]. It was shown the feasibility of the hybrid system for the production of energy in an electric vehicle. Mezzai et al. examined a hybrid PV/wind/FC power system and simulated the system under Matlab/simulink [25]. The development of the mathematical model topology and the power management of the global system with battery bank system were the significant contributions of their research. Eid proposed a modeling, control and performance of a PV/wind/FC hybrid system under variable load demands and different environmental conditions [26]. The system consisted of 60 kW wind turbine energy conversion system, 40 kW PV panel, 40 kW FC module, DC and AC loads. Bigdeli compared optimal hybrid approaches including PV, FC and battery [27]. He examined fuzzy logic controller and observation methods to achieve maximum power point tracking for the PV system. He also proposed an intelligent hybrid system management which could use PV and FC according to the system demand intelligently, by applying some optimization algorithms. Dash and Bajpai formulated a power management strategy for the integration of the power output from PV array, FC stack and battery with a provision for onsite hydrogen production by means of an electrolyzer and Hydrogen tank [28]. It was concluded by the authors that the control strategy handled the source power effectively by considering the limited lifecycle of storage devices. It also eliminated the need for a dump load in the system by matching the source power with load demand when the storage devices were charged to their maximum capacity. Bensmail et al. studied a hybrid PV/FC for stand-alone applications [29]. They produced Hydrogen which was used by the FC stack to produce electrical energy to supply the Direct Current bus. Maleki and Pourfayaz modeled, sized and performed a cost analysis of a PV/wind/diesel/battery/FC by using the harmony search algorithm [30]. The investigators compared two scenarios of hybrid systems namely PV/wind/diesel/battery and PV/wind/diesel/FC. Their numerical results indicated that the conventional PV/Wind/diesel/battery system was more cost-effective than PV/Wind/diesel/FC system. Nevertheless, using PV/Wind/diesel/FC energy source led to a reliable energy source and with improvement in the efficiency of both FC and electrolyzer, it was demonstrated that the FC/electrolyzer storage system could be economically competitive. Ren et al. developed a PV/FC/Battery hybrid energy system for residential applications [31]. The authors noticed that the hybrid system satisfied electric and thermal demands. It was also shown that annual operation cost and CO₂ emissions were minimized. Isa et al. assessed the viability to develop a PV/FC/Battery system to meet the load demand of a hospital building in Malaysia [32]. The simulation result demonstrated that the proposed cogeneration system had the lowest net present cost (NPC), LCOE and operating cost which were \$ 106,551, 0.091 \$/kWh and 7245 \$/yr respectively. Gadalla and Zafar investigated and analyzed a Hydrogen FC-PV power system for small unmanned aerial vehicle [33]. The endurance of the unmanned aerial vehicle increased 100% with the use of hybrid hydrogen FC-PV power system. The endurance increased to 53,396 s with a hybrid system, as compared to 28,173 s with just the fuel cell. Heydari and Askarzadeh performed a techno-economic analysis of a PV/Biomass/FC energy system taking into account different FC system initial capital costs (2000 \$/kW to 4000 \$/kW) and the amount of the FC generation from the total generation of the hybrid system was evaluated by considering loss of power supply probability [34]. Ezzat and Dincer developed, analyzed and assessed a PV/FC/Li-ion Battery system powered vehicle [35]. They showed that the overall energy and exergy efficiencies became 39.46% and 56.3%, respectively at a current density of 1150 mA/cm² for FC-battery. Moreover, energy and exergy efficiencies were found to be 39.86% and 56.63% at current density of 1150 mA/cm² for FC-battery-PV. Tiar et al. proposed a fuzzy logic-based power management of a PV/FC hybrid power system [36]. The proposed power manager allowed the system to switch smoothly between the described operating modes, respecting the stack capability. Das et al. performed a feasibility analysis of a PV/Battery/FC hybrid energy

system for an indigeneous residence in East Malaysia [37]. The optimization and sensitivity analysis using HOMER showed that the PV/Battery system was economical with a NPC and COE of \$ 335,297 and 0.323 \$/kWh with no emission. The performance improvement of a Battery/PV/FC/grid hybrid energy system considering load uncertainty modeling using information gap decision theory was examined by Nojavan et al. [38]. The sensitivity analysis revealed that the PV/Battery system was economical and environmentally friendly with total net present cost of \$ 335,297 and cost of energy of 0.323 \$/kWh with no emission. Majidi et al. proposed a multi-objective model for optimal operation of a PV/Battery/FC/grid hybrid energy system using weighted sum technique and fuzzy satisfying approach considering responsible load management [39]. It was noticed that optimal cost emission operation of PV-battery-FC hybrid system was obtained by demand response program implementation. Nojavan et al. evaluated the optimal performance of an on-grid PV/FC/Battery hybrid system in the presence of demand response program with considering electrical load uncertainty [40]. The results allowed to conclude that due to the implementation of demand response program, load was shifted from peak periods to offpeak periods and as a result, total cost of hybrid system was reduced of 2.97%. Singh et al. performed a techno-economic feasibility analysis of Hydrogen FC and PV hybrid renewable energy system for academic research building [41]. The results indicated that the Hydrogen FC and battery bank were the most significant modules of the hybrid system to meet load demand at late night and early morning hours. A multi-objective model was proposed for economic and environmental operation of a battery/FC/PV hybrid energy system in the presence of demand response program [42]. It was revealed from the obtained results that due to positive effects of demand response program, the economic and environmental ideals were achieved. A techno-economic analysis based on integrated modeling, simulation, and optimization approach was used to design an off grid hybrid PV/FC power system [43]. The simulation results showed that the distributed power generation using solar PV and FC energy systems integrated with an electrolyzer for Hydrogen production and using cycle charging dispatch control strategy presented the best performance. Han et al. provided a two-level energy management strategy for PV/FC/Battery-based DC Microgrid [44]. The analysis and comparison of the results showed that the proposed two-level energy management strategy could achieve lower equivalent hydrogen consumption than classical Proportional-Integral control and state machine control method. Sadeghi and Askari presented a prefeasibility techno-economic assessment of a hybrid power plant with PV, FC and compressed air energy storage used to supply the electricity needs of 500 households with peak electricity demand of 500 kW [45]. The results revealed that the overall system efficiency would be increased by about 25%, when the compressed air energy storage was used and the compressor switched off. Samy et al. developed a flower pollination optimization algorithm for an off-grid PV/FC hybrid renewable system [46]. The optimum system configuration was acquired using flower pollination optimization with the optimal hybridization of 27 solar PV, 28 FCs, 58 electrolyzers and 37 Hydrogen storage tanks for a loss of power supply probability of 1.52%. Ghenai and Bettayeb modeled and analyzed the performance of a stand-alone hybrid PV/FC/Diesel generator power system for university building [47]. The proposed stand-alone hybrid renewable power system had high renewable fraction (66.1%), was economically feasible (92 \$/MWh) and environmentally friendly (24 kg CO₂/MWh). Jamshidi and Askarzadeh proposed a techno-economic analysis and size-optimization of an off-grid hybrid PV/FC/Diesel generator [48]. Based on the results, it was noticed that the integration of Hydrogen energy technology reduced the total cost of the hybrid energy systems. An adaptive multi-input and single-output fuzzy controller which was designed in a supervisory manner for a multi-agent system was proposed by Yu et al. [49]. Ghenai et al. presented an optimal design and performance analysis of a hybrid PV/FC/Diesel generator power system for cruise ship in Stockholm in Sweden [50].

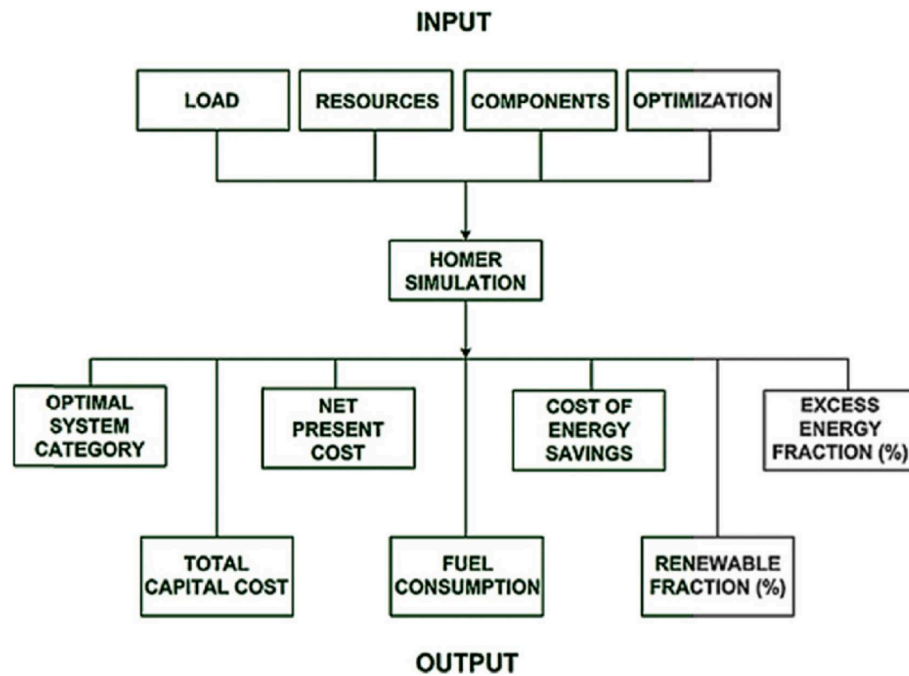


Fig. 1. Architecture of HOMER Pro software [64].

The proposed renewable energy system offered a good penetration of renewable energy system (13.83%), and greenhouse gas and particular emissions reduction (9.84% emissions reduction compared to baseline system using Diesel engines). Budak and Devrim performed a comparative study of PV/FC hybrid energy system based on methanol and water electrolysis [51]. They showed that the use of the methanol electrolyzer could produce 27% more Hydrogen than the water electrolyzer. Ozgürin et al. modeled and simulated a hybrid PV-Electrolyzer-FC for micro-cogeneration applications [52]. The study showed that this hybrid power system provided a viable option for powering stand-alone house in a self-sustained manner. Samy et al. addressed the problem of power outages in distant districts of Egypt by proposing a PV/wind turbine/FC system [53]. Simulation results showed that the optimal system for solving the grid unavailability consisted of eighty PVs, two WTs, twenty FCs, forty-one electrolyzers, and one hundred eighteen hydrogen tanks. Okundamiya described the size optimization of a hybrid PV/FC grid linked power system including hydrogen storage in Nigeria [54]. He demonstrated that a hybrid system comprising 54.7 kW photovoltaic array, 7 kW fuel cell system, 14 kW power inverter and 3 kW electrolyzer with 8 kg hydrogen storage tank could sustainably augment the erratic grid with a very high renewable fraction of 96.7% at \$0.0418/kWh. Yilanci et al. proposed a review on solar hydrogen production methods [55]. They also discussed about solar-Hydrogen/Fuel cell hybrid systems for stationary applications. Various methods based on solar-Hydrogen were presented in their paper. Indeed, they highlighted four different methods for solar-Hydrogen production. The first method was the solar-thermal Hydrogen production divided into two categories: thermolysis of water which needs extremely high temperatures, and thermochemical production based on the use of concentrated solar radiation as the energy source for performing high-temperature reactions [56–60]. The second method was the Hydrogen production via solar electricity which takes place via the photovoltaic cells. Here, the electrolysis of water could be carried out by a current generated by the photovoltaic cells [61]. The third approach was the photoelectrolysis of water based on the conversion of solar light to clean, storable chemical fuel Hydrogen. It uses photoelectrochemical light collecting systems to power the electrolysis of water [62]. The last method was the photobiological Hydrogen production. It uses the same processes found in plant and algal photosynthesis, but it adapts these

processes for the generation of Hydrogen gas instead of carbon containing biomass [63]. The authors also considered three different energy demand paths, from PV panels to the consumer. Minimum and maximum overall energy and exergy efficiencies of the system were calculated based on these paths. It was found that the overall energy efficiency values of the system varied between 0.88% and 9.7%, while minimum and maximum overall exergy efficiency values of the system were between 0.77% and 9.3% as a result of selecting various energy paths.

It appeared from this literature review that hybrid systems based on the combination of photovoltaics panels, electrolyzer and Hydrogen fuel cells were widely used to overcome the intermittency problem inherent with PV, and increased the sustainability of the system by considering the environmental aspect. Unfortunately, it is worth mentioning that such hybrid systems were not yet implemented in some developing countries (Cameroon for instance), although there are many electrification projects based on the use of hybrid renewable energy systems. This paper therefore investigates for the very first time the techno-economic feasibility by using HOMER Pro of two scenarios of hybrid systems namely, PV/Fuel Cell/Electrolyzer/Biogas, and PV/Battery/Fuel Cell/Electrolyzer/Biogas for energy and Hydrogen production in the city of Maroua, recognized as being part of the sunniest region (Far North) of Cameroon. The combination of electrolyzer, fuel cell and hydrogen tank was used in the present design to reduce battery storage requirement. Three types of household electricity demands communities (low, medium and high consumers) were considered in this work. The rest of the paper is organized as follows: Section 2 presents the materials and methods, section 3 shows the results and discussions and concluding remarks are drawn in section 4.

2. Materials and methods

In this study, HOMER Pro software was used for the sizing and economic performance of two scenarios of hybrid systems namely, PV/Fuel Cell/Electrolyzer/Biogas, and PV/Battery/Fuel Cell/Electrolyzer/Biogas. This software has numerous energy components. Simulation, optimization and sensitivity analysis can be performed. The architecture of HOMER Pro software is shown in Fig. 1 [64].



Fig. 2. Map of Cameroon and the location of the city of Maroua [65].

2.1. Study location

The city of Maroua (latitude: 10° 35' North, and longitude: 14° 19' East) is the chief town of the Far North region and the Diamaré division. It has a population of about 1,007,000 inhabitants. The climate is tropical, dry and hot, almost semi-desert. The average annual temperature is 28.3 °C and the annual precipitation is 794 mm³. The location of the city of Maroua is presented in Fig. 2 [65]. It records the highest solar radiation in the country with about 5.80 kWh/m²/year [66].

2.2. Load profile estimation

It is important to mention that our approach consisted of data collection from electricity users (households, businesses, municipalities, schools, etc.) across the city of Maroua surveyed. Three categories of household electricity demand communities (low, medium and high consumers) were considered in this work in order to evaluate the monthly load profile. A low consumer is defined as a consumer that depends mostly on farming. The electrical energy is used here particularly for lighting. A peak load of 110 W was recorded in the evening. A medium consumer, apart from the usage of the low consumer, uses an electric iron and a water heater in the morning for their breakfast, a laptop in the evening, a television in the afternoon and in the evening. A peak load of 170 W was observed in the evening. High consumer of electricity has the usage of medium consumer but additionally, the high consumer has an air conditioner (operating between 12 a.m. and 16 a.m.) and a refrigerator which was regulated by a thermostat for energy saving. A peak load of 1520 W was observed in the evening. The daily load profile approximation of the different categories of communities is presented in Table 1. In this work, 200 households of each community were taken into consideration. Figs. 3–5 present the daily load profile approximation for a community of 200 high, medium and low consumers respectively.

Table 1
Daily load profile approximation.

Consumer	Appliance	Power (W)	Quantity	Number of hours per day	Total power (W)	Energy (Wh)/Day
Low-	Phone charger	5	3	8.33	15	125
	Radio	15	1	8	15	120
	Security bulb	15	1	11	15	165
	Lighting bulbs	15	5	5	75	375
	Random load	15	1	5	15	75
	Total					860
Medium-	Phone charger	5	3	9	15	135
	Radio	15	1	11	15	165
	Security bulb	15	1	11	15	165
	Lighting bulbs	15	5	5	75	375
	Water heater	1000	1	0.125	1000	125
	Electric iron	1000	1	0.125	1000	125
	TV	36	1	6.94	36	250
	Laptop	15	1	3	15	45
	Random load	15	1	5	15	75
	Total					1460
High-	Phone charger	5	3	9	15	135
	Radio	15	1	11	15	165
	Security bulb	15	1	11	15	165
	Lighting bulbs	15	5	5	75	375
	Water heater	1000	1	0.125	1000	125
	Electric iron	1000	1	0.125	1000	125
	TV	36	1	8	36	288
	Laptop	15	1	3	15	45
	Refrigerator	250	1	8	250	2000
	Air conditioner	1629.25	1	4	1629.25	6517
	Random load	15	1	4	15	60
	Total					9535

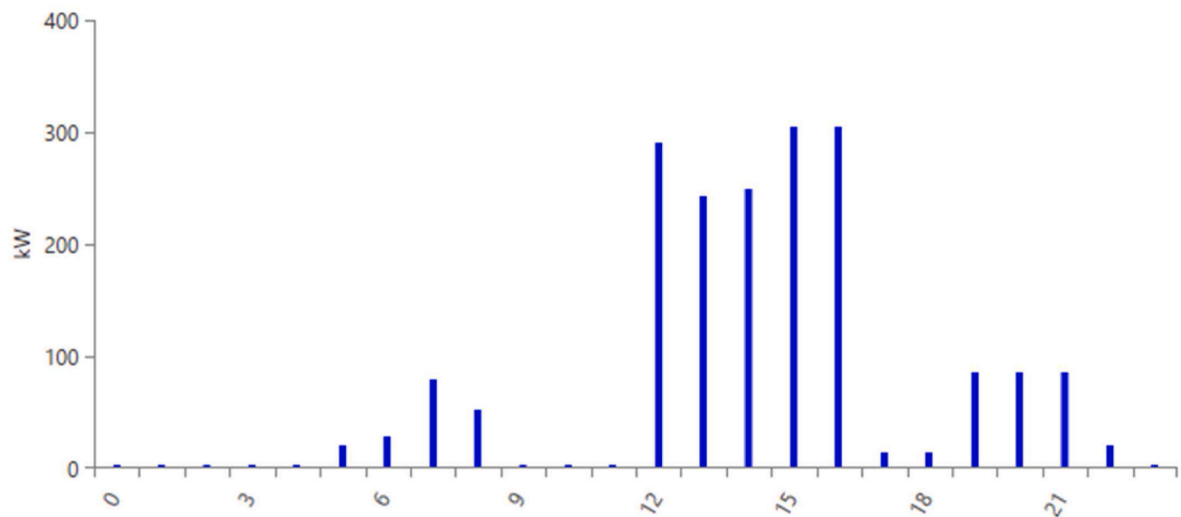


Fig. 3. Daily load profile estimation for a community of 200 high consumers.

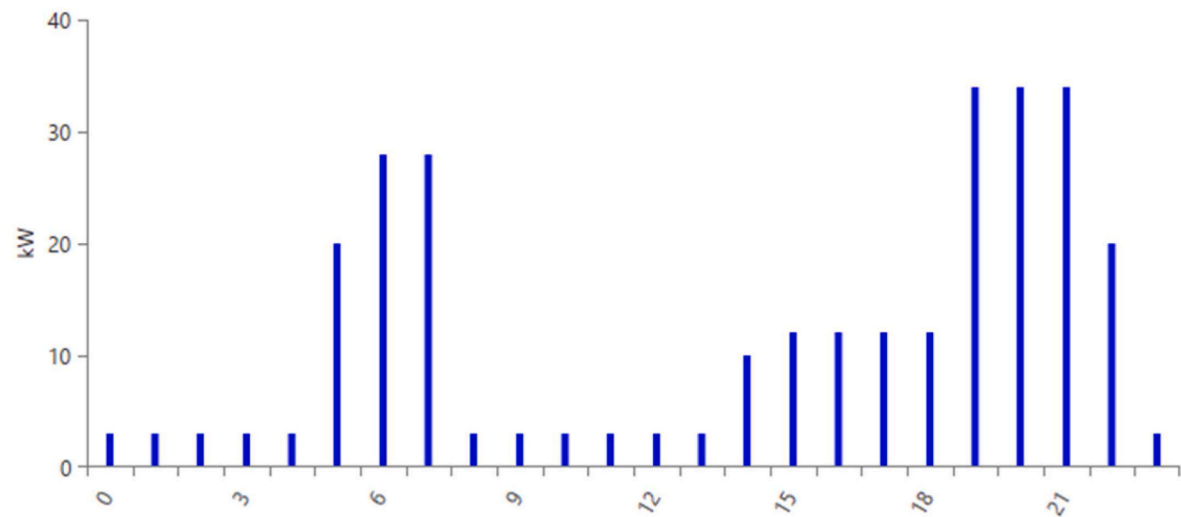


Fig. 4. Daily load profile estimation for a community of 200 medium consumers.

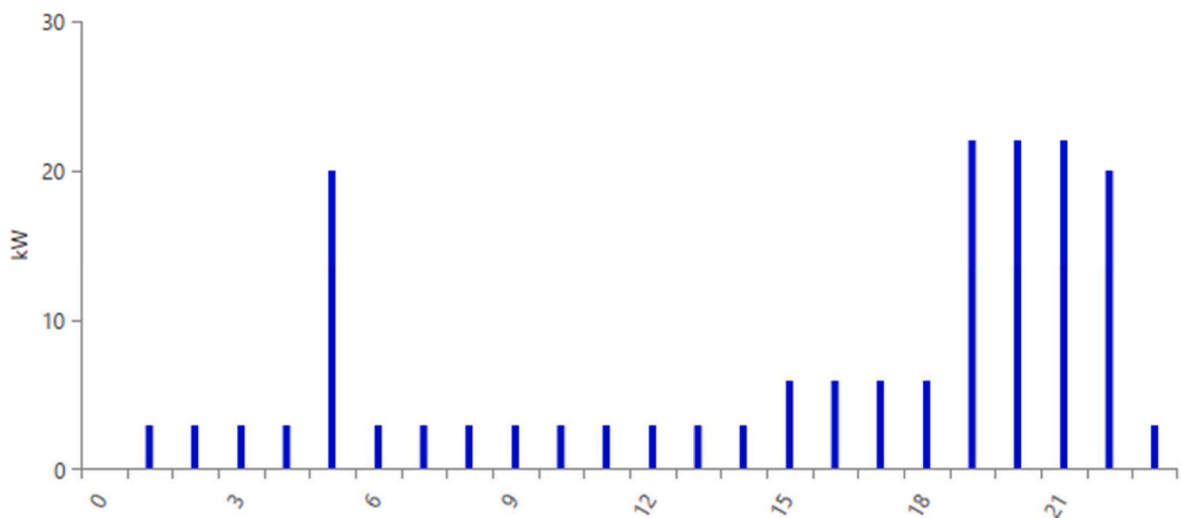


Fig. 5. Daily load profile estimation for a community of 200 low consumers.

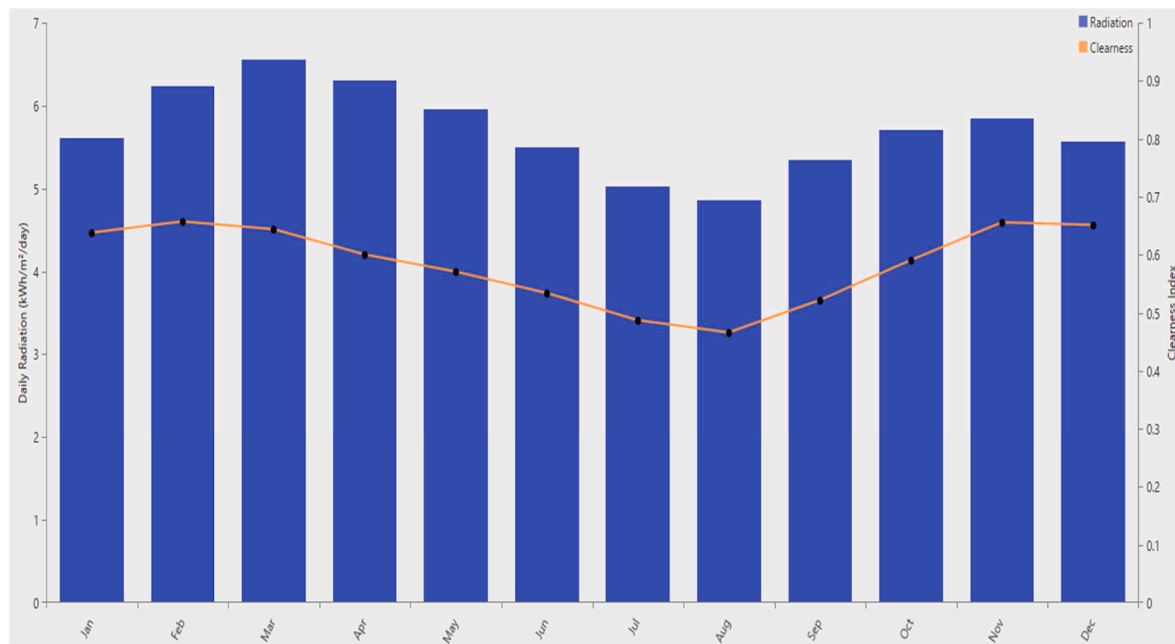


Fig. 6. Solar resources of the city of Maroua, Cameroon.

Table 2

Electricity potential from biomass from Maroua.

Livestock	Number	Dung availability (kg/head/day)	Total dung (kg/day)	Total dung (recovery factor = 0.7)	Total biogas yield (m ³ /day)	Potential power yield (kWh/day)
Sheep	800	1	800	560	20.16	27.60
Goats	725	1	725	507.5	18.27	25.03
Cows	900	10	9000	6300	226.80	310.68
Total	2425	–	10,525	7367.5	265.23	363.31

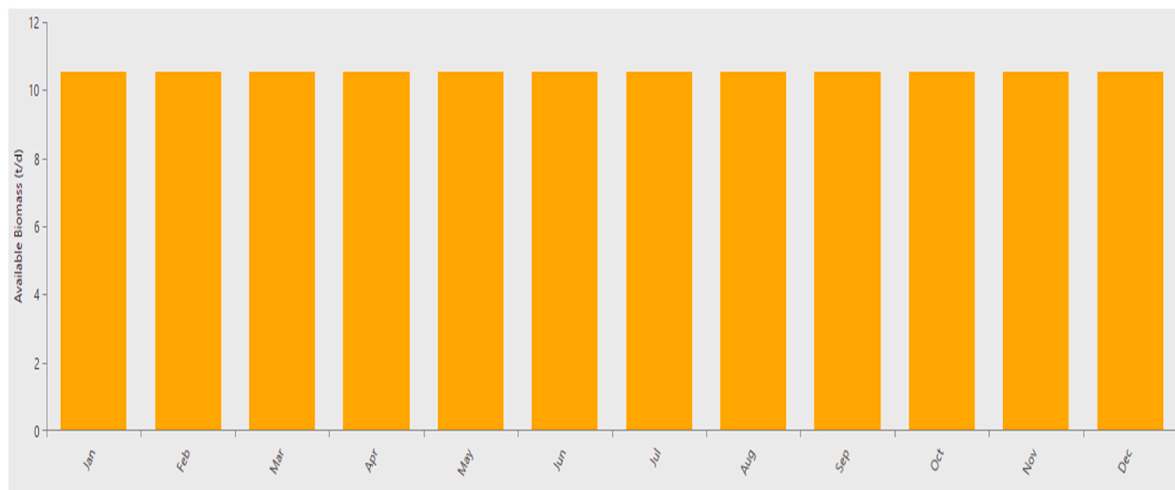


Fig. 7. Biomass (livestock dung) technical potential for the city of Maroua, Cameroon.

2.3. Solar and biomass resources

2.3.1. Solar resources

The solar resources of the city of Maroua were collected from the NASA surface meteorology and solar energy database. It was observed an annual average radiation of 5.71 kWh/m²/day, with an average clearness index of 0.584 (this fraction means more than half of the solar radiation is striking the surface of the earth). It can be seen from Fig. 6

that the maximum and minimum solar radiations occurred in March (6.56 kWh/m²/day) and August (4.85 kWh/m²/day) respectively. It was concluded that in March, the highest production capacity of photovoltaic panels could be expected. The monthly solar radiation and clearness index of the city of Maroua are presented in Fig. 6.

2.3.2. Biomass resources

Biomass represents all organic matter that can be transformed into

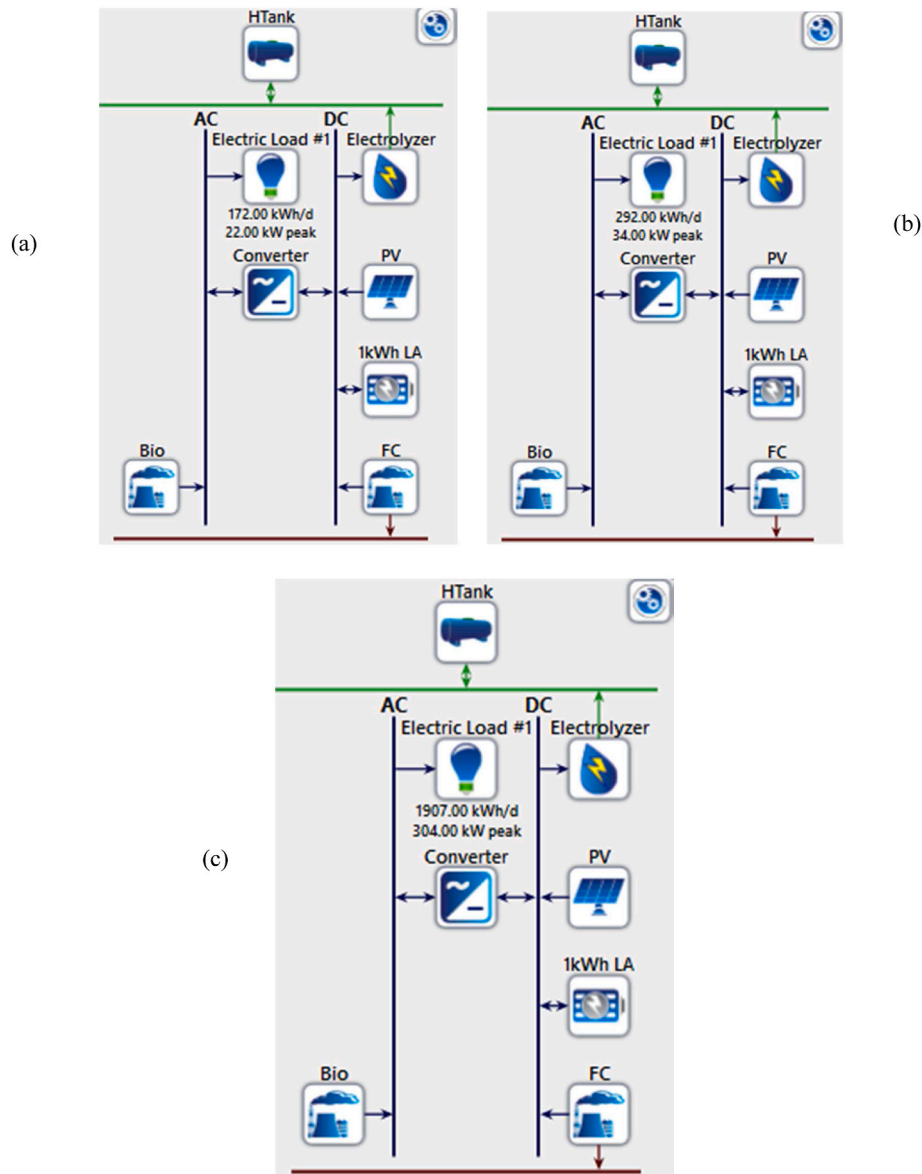


Fig. 8. Design of PV/Battery/Fuel Cell/Electrolyzer/Biogas hybrid system in HOMER Pro for the three types of communities: low consumers (a), medium consumers (b) and high consumers (c).

energy. Energy recovery from biomass can produce heat or electricity. Dung generated by the intense livestock activity in the Far North region were considered for biogas production. An average population of livestock made up of 800 sheeps, 725 goats and 900 cows was used. It was assumed that a cow produces 10 kgs of dung per day while a goat or a sheep produces 1 kg per day [67]. The following assumptions were also made [68]:

- A recovery factor of 0.7;
- A production of 0.036 m^3 per kg of wet dung;
- A production of 1 kWh for 0.73 m^3 of biogas.

Table 2 shows the electricity potential from biomass for the site of Maroua. Fig. 7 shows the biomass (animal dung) technical potential in Maroua. In this study, the gasification ratio (amount of biogas produced per unit mass of biomass), the lower heating value (amount of energy contained in 1 kg of biogas available to feed the biogas generator), the cost of biomass were taken as 0.05 kg/kg, 5.5 MJ/kg and 0 \$/t respectively. Concerning the dung recovery in the present study, it was considered farms with feed provided for animals kept in farm buildings.

The cost of the fuel was considered to be the cost associated to the collection of dung in the farm buildings.

2.4. Modeling using HOMER Pro

The modeling of two scenarios of hybrid systems namely, PV/Fuel Cell/Electrolyzer/Biogas, and PV/Battery/Fuel Cell/Electrolyzer/Biogas was performed by using HOMER Pro in this paper. Different components were used for the modeling, namely: PV panels, a biogas generator, batteries, converters, an electrolyzer, a fuel cell and a Hydrogen tank. The designs of PV/Battery/Fuel Cell/Electrolyzer/Biogas, and PV/Fuel Cell/Electrolyzer/Biogas hybrid systems for the different communities are presented in Figs. 8 and 9. The details of the specifications of the various components are also given in this section.

2.4.1. PV modules

The power production of a PV array can be expressed as [77]:

$$P_{PV} = n_{PV} U_{PV} i_{PV} \quad (1)$$

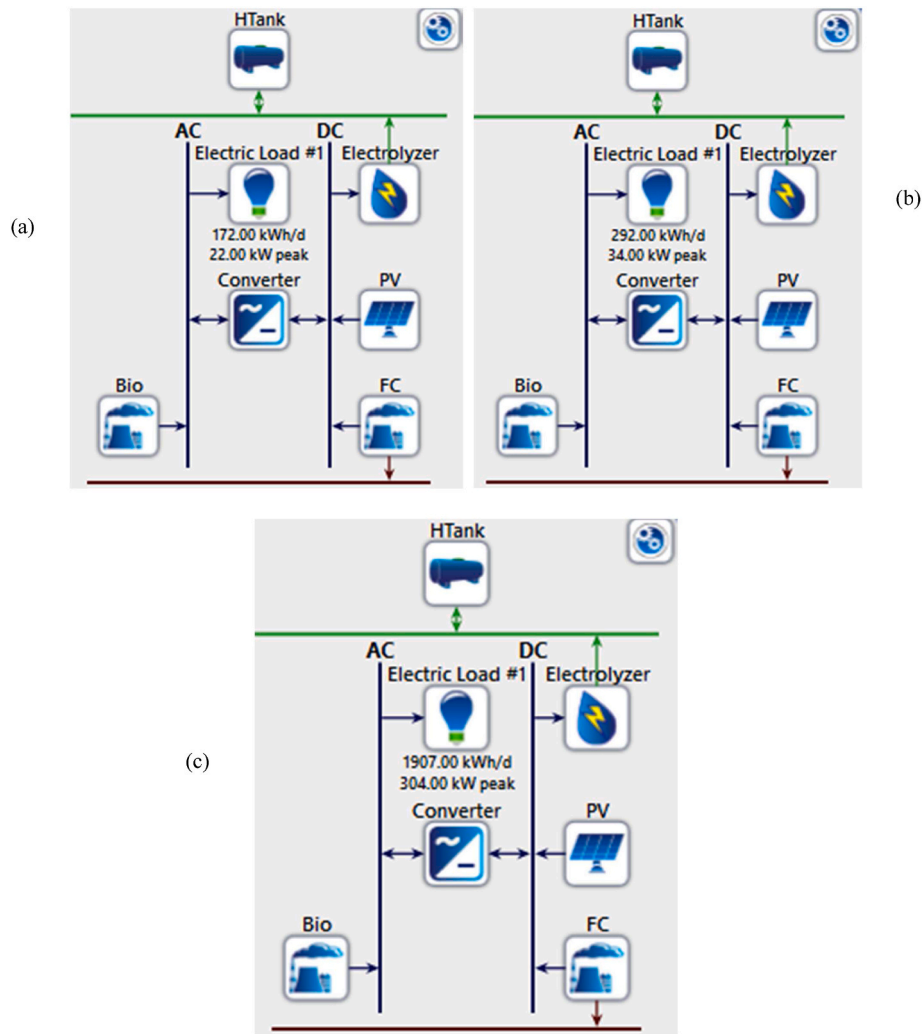


Fig. 9. Design of PV/Fuel Cell/Electrolyzer/Biogas hybrid system in HOMER Pro for the three types of communities: low consumers (a), medium consumers (b) and high consumers (c).

Table 3
Specifications of the technical parameters for the PV modules [69].

Model	BP350
Maximum power (W)	50
Nominal voltage (V)	12
Maximum power voltage (V)	17.5
Maximum power current (A)	2.9
Open-circuit voltage (V)	21.8
Short-circuit current (A)	3.17
Surface (m ²)	0.451
Capital cost (\$)	148.5
Replacement cost (\$)	133.65
Operation and maintenance cost (\$)	14.85

Table 4
Specifications of the technical parameters for the converter [70].

Capital cost (\$)	985
Replacement cost (\$)	887
Operation and maintenance cost (\$)	98
Lifetime (years)	20
Efficiency (%)	95
Relative capacity	80%

Where n_{PV} , U_{PV} , and i_{PV} are the number of PV modules, the voltage and the current of the PV module. Concerning the optimum operating voltage and point current, they are given by Ref. [77]:

$$\begin{cases} U_{PV} = U_{max} \left[1 + 0.0539 \log \left(\frac{I_T}{I_{st}} \right) \right] + \beta_0 (T_A + 0.02 I_T - T_{st}) \\ i_{PV} = i_{sc} \left(1 - \left(1 - \frac{i_{max}}{i_{sc}} \right) e^{\left(\frac{U_{max}}{K_2 U_{OC}} \right)} \left[e^{\left(\frac{U_{max}}{K_2 U_{OC}} \right)} - 1 \right] \right) + \Delta i \end{cases} \quad (2)$$

with

$$K_2 = \frac{U_{max}}{\ln \left(1 - \frac{i_{max}}{i_{sc}} \right)} \quad (3)$$

$$\Delta i = \alpha_0 \left(\frac{I_T}{I_{st}} \right) (T_{cell} - T_{st}) + \left(\frac{I_T}{I_{st}} - 1 \right) i_{sc} \quad (4)$$

Here, i_{sc} , U_{max} , U_{OC} , i_{max} , α_0 , β_0 , I_T , I_{st} , T_{st} and T_A are the short-circuit current (in Ampere), the maximum power voltage (in Volts), the open-circuit voltage (in Volts), the maximum power current (in Watts), the current temperature coefficient, the voltage temperature coefficient of the module, the total solar radiation on tilted panel (in W/m^2), the standard solar radiation (1000 W/m^2), the standard ambient temperature (25⁰ C) and the ambient temperature (in Celcius). BP350 solar

Table 5
Specifications of the technical parameters for the batteries [71].

Nominal capacity (Ah)	253
Voltage (V)	12
Minimum charge (%)	20
Capital cost (\$)	465
Replacement cost (\$)	418.5
Operation and maintenance cost (\$)	46.5

Table 6
Biogas generator properties and specifications parameters [72].

Capacity	15 kW
Capital cost (\$)	2000
Replacement cost (\$)	1800
Operation and maintenance cost (\$)	200
Lifetime (hours)	20,000
Fuel price (\$/kg)	0.053
Sulfur content (%)	0
Carbon content (%)	5
Density (kg/m ³)	0.720
Lower heating value (MJ/kg)	5.5
Minimum load ratio (%)	30

panels (50 W) were used in this study. The specifications of PV modules used in the design are shown in Table 3 [69].

2.4.2. Converter

A SUNGOLDPOWER UL1741 converter was incorporated in the design. The technical parameters for the converter are shown in Table 4 [70].

2.4.3. Batteries

The battery bank capacity can be expressed as a function of time through the following equation [78]:

$$\begin{cases} C_{B\text{charg}}(t) = C_{B\text{charg}}(t-1)(1-\sigma) + \left(P_T(t) - \frac{P_L(t)}{\eta_{\text{conv}}} \right) \eta_{\text{Batt}} \\ C_{B\text{discharg}}(t) = C_{B\text{discharg}}(t-1)(1-\sigma) + \left(\frac{P_L(t)}{\eta_{\text{conv}}} - P_T(t) \right) \eta_{\text{Batt}} \end{cases} \quad (5)$$

With $C_{B\text{charg}}(t)$ the capacity of the battery bank during the charging mode, and $C_{B\text{discharg}}(t)$ the capacity of the battery bank during the discharge mode. $P_T(t)$ and $P_L(t)$ are the total output power generated by the system and the load demand at the corresponding hour respectively. η_{Batt} and η_{conv} are the battery efficiency and the conversion efficiency.

The Concorde (PVX-2120L) batteries were added in the system. The specifications of the batteries integrated are listed in Table 5 [71].

2.4.4. Biogas generator

The properties of the biogas generator and the specifications parameters are shown in Table 6 [72].

2.4.5. Fuel cell

Proton Exchange membrane fuel cell was considered for the conversion of the chemical energy to electricity through electrochemical

Table 7
Fuel cell properties and specifications parameters [73].

Initial capital (\$/kW)	3000
Replacement (\$/kW)	2700
Operation and maintenance cost (\$/kW)	300
Lifetime (hours)	50,000
DC power consumption	3.9–4.4 kWh/Nm ³
Dionized water consumption	900 L/h
Hydrogen production	0–1000 Nm ³ /h

Table 8
Specifications parameters for the Electrolyzer [74].

Initial capital (\$/kW)	2000
Replacement (\$/kW)	1800
Operation and maintenance cost (\$/kW)	200
Lifetime (years)	15

Table 9
Specifications parameters for the Hydrogen tank [75].

Initial capital (\$/kg)	1500
Replacement (\$/kg)	1350
Operation and maintenance cost (\$)	150
Lifetime (years)	15

reaction. Indeed, it received Hydrogen and converted it into electricity, heat and water with no pollutants. The properties of the fuel cell and its specifications parameters are presented in Table 7 [73].

2.4.6. Electrolyzer

The electrolyzer is the central element for the Hydrogen production. The water electrolyzer was used in the present design. The specifications parameters implemented are shown in Table 8 [74].

2.4.7. Hydrogen tank

The Hydrogen produced by the electrolyzer was stored in a 30 bar Hydrogen tank. The specifications parameters used are shown in Table 9 [75].

2.5. Simulations and optimization using HOMER Pro

2.5.1. Assessment criteria

For the simulations and optimization procedures, the Net Present Cost (NPC) and the levelized cost of energy (LCOE) were used as assessment criteria for the hybrid system optimization. These assessment criteria are given as [59]:

$$\begin{cases} NPC = \frac{C_{\text{ann,tot}}}{CRF(d, N)} \\ C_{\text{ann,tot}} = C_{\text{ann,cap}} + C_{\text{ann,rep}} + C_{\text{ann,O\&M}} + C_{\text{ann,fuel}} - R_{\text{ann,salv}} \\ CRF = \frac{d(1+d)^N}{(1+d)^N - 1} \\ d = \frac{i-f}{1+f} \\ COE = \frac{C_{\text{ann,tot}} - C_{\text{boiler}} \bullet H_{\text{thermal}}}{E_{\text{served}}} \end{cases} \quad (6)$$

Where $C_{\text{ann,tot}}$, $C_{\text{ann,cap}}$, $C_{\text{ann,rep}}$, $C_{\text{ann,O\&M}}$, $C_{\text{ann,fuel}}$, $R_{\text{ann,salv}}$, CRF , d , N , i , and f are total annualized cost of the system (\$/year), annualized capital (\$/year), replacement (\$/year), maintenance costs of all components of the system (\$/year), annualized cost of fuels used to feed the power generators (\$/year), annualized total salvage value of all system components (\$/year), capital recovery factor which is a function of annual real interest rate (8%) [76], discount rate, project lifespan, nominal interest rate and annual inflation respectively. COE , C_{boiler} , H_{thermal} and E_{served} are levelized cost of energy, boiler marginal cost (\$/kWh), total thermal load served (kWh/year) and the total electric load served (kWh/year). By considering that there is no thermal load served in the system, $H_{\text{thermal}} = 0$. In this paper, the discount and inflation rates of 5% and 3% were considered respectively [76].

2.5.2. Dispatch strategy

Here, two dispatch strategies were modeled in HOMER Pro, namely:

Table 10
Sensitivity variables and their values.

Sensitivity variable	Values
Nominal discount rate (%)	{8; 9.20; 10.40; 11.60; 12.80; 14.00}
Project lifetime (years)	{15; 17; 19; 21; 23; 25}

load following (LF) and cycle charging (CC) [59].

2.5.3. Sensitivity analysis

The discount rate and project lifetime were used as sensitivity variables in the simulations. Table 10 presents the sensitivity variables and their values.

3. Results and discussions

3.1. Optimization and sensitivity results

First of all, it is important to summarize the calculation report in HOMER Pro for the different categories of electricity demands communities (see Table 11). It was observed that an average of 82.38% of the simulated solutions were feasible. It was also noticed that adding the batteries in the design of all the categories of electricity demands communities increased the feasibility of the solutions. It was concluded that the systems architectures were relevant. Table 12 shows the best hybrid system configuration for all the categories of household electricity demand communities. In Table 12, scenario 1 was the PV/FC/Electrolyzer/Biogas system and scenario 2 the PV/Battery/FC/Electrolyzer/Biogas system.

Table 12 shows the optimization results of the candidate winning system which is the first solution of the output HOMER calculations for each scenario. The results meet the following sensitivity analysis parameters: solar scaled annual average irradiation of 5.75 kWh/m²/day,

nominal discount and expected inflation rates of 8 and 3%, project lifetime of 15 years, maximum capacity shortage of 10, and biomass scaled annual average of 10.53 ton/day. The first columns showed the architecture of the system with the associated dispatched strategy; the various costs (NPC, LCOE and LCOH) and the power production of each component of the architecture. Generally, the optimization results revealed solutions that included PV and biogas generators. In the case that the solution involved the combination of fuel cell, electrolyzer and Hydrogen tank (scenario 1), the system's LCOE, NPC and LCOH increased when compared to scenario 2. It was demonstrated that the optimized systems consisted of systems that had electrolyzer, fuel cell, hydrogen tank combination that worked as a storage system replacing the battery bank (scenario 1) and systems that had battery bank as storage system (scenario 2). It was noticed that the scenario 2 had the minimum cost of energy with 0.139, 0.091 and 0.071 US\$/kWh for low, medium and high consumers respectively. It had the least Hydrogen production costs which were 3.06, 1.34 and 0.15 US\$/kg for low, medium and high consumers respectively. In addition, it was shown that the total net present cost was higher for the systems incorporating the electrolyzer, fuel cell, hydrogen tank combination as the storage system. The results also presented the conclusion that there was no unmet load for all the configurations considered. It was also concluded that the PV modules contributed largely for the power production followed by fuel cell for the scenario 1. For scenario 2, the PV modules were followed by the biogas generator in terms of electricity generation. The Hydrogen tank capacity was 5000 kgs for low and medium consumers while that of high consumers was 25,000 kgs.

Fig. 10 depicts the monthly average electricity production for optimal hybrid solutions generated from the biogas generators, solar panels, and fuel cell for each type of household electricity demand community in scenario 1. It was observed that the component with the highest annual electricity production was the PV array module (92.27%). It was followed by the biogas generator (5.43%) and the fuel

Table 11
Calculation report in HOMER Pro for the different categories of electricity demands communities.

Category	Scenario	Number of solutions simulated	Number of feasible solutions	Percentage (%)
Low consumers	PV/FC/Electrolyzer/Biogas	14,442	12,672	87.74
	PV/Battery/FC/Electrolyzer/Biogas	137,808	135,306	98.18
Medium consumers	PV/FC/Electrolyzer/Biogas	29,304	26,550	90.60
	PV/Battery/FC/Electrolyzer/Biogas	127,512	116,946	91.71
High consumers	PV/FC/Electrolyzer/Biogas	50,544	28,626	56.63
	PV/Battery/FC/Electrolyzer/Biogas	147,552	103,155	69.91
Average				82.38

Table 12
Optimization results.

Item	Component	Unit	Best hybrid system					
			Scenario 1			Scenario 2		
			Low-consumer	Medium-consumer	High-consumer	Low-consumer	Medium-consumer	High-consumer
System architecture	PV array	kW	144	879	11,925	138	234	820
	Biogas generator	kW	15	15	15	15	15	15
	Converter	kW	11	31.9	570	27.2	57.8	405
	Batteries	Quantity	–	–	–	480	1023	9519
	Fuel cell	kW	15	24	266	15	24	266
	Electrolyzer	kW	15	24	266	15	24	266
	Hydrogen tank	kg	5000	5000	25,000	5000	5000	25,000
	Dispatch strategy	LF or CC	CC	CC	CC	CC	LF	CC
Cost	LCOE	US\$/kWh	0.871	0.898	1.524	0.139	0.091	0.071
	NPC	US\$	98.4 M	172 M	1.91B	15.7 M	17.5 M	88.7 M
	LCOH	US\$/kg	7.66	4.95	0.45	3.06	1.34	0.15
Power production	PV array	kWh/year	248,324	1,514,060	20,540,570	238,391	403,749	1,412,128
	Biogas generator	kWh/year	42,885	33,329	10,091	165	34.5	0
	Fuel cell	kWh/year	11,911	37,468	121,553	0	0	0
	Total	kWh/year	303,119	1,584,858	20,672,214	238,556	403,784	1,412,128
	Unmet load	kWh/year	0	0	0	0	0	0

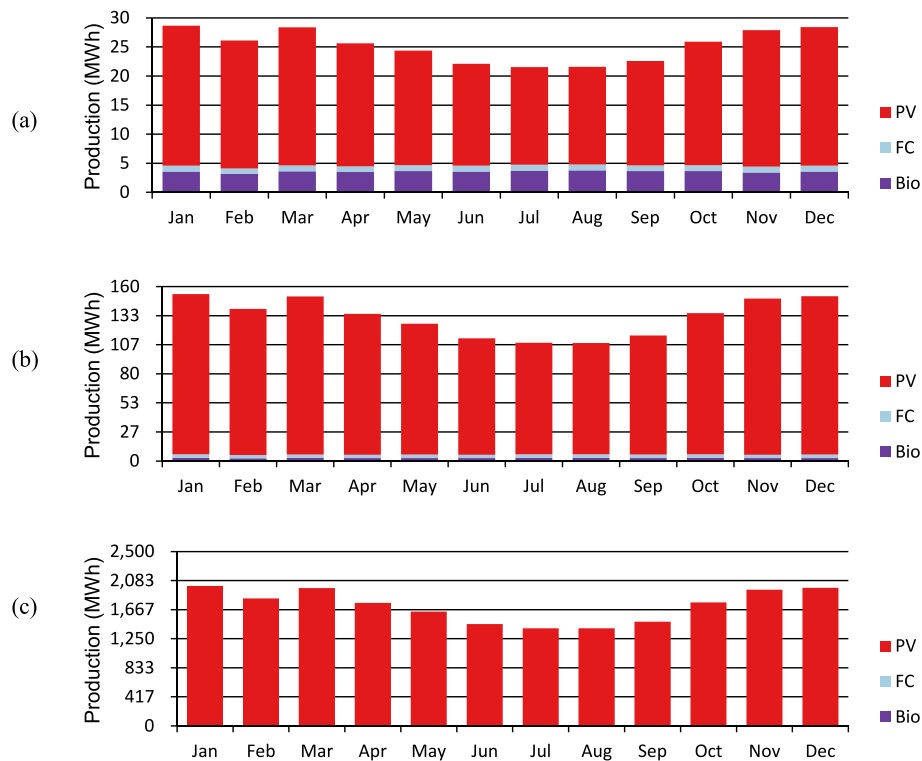


Fig. 10. Monthly average electricity production for optimal hybrid solutions for low-consumers (a), medium consumers (b) and high consumers (c) communities in scenario 1.

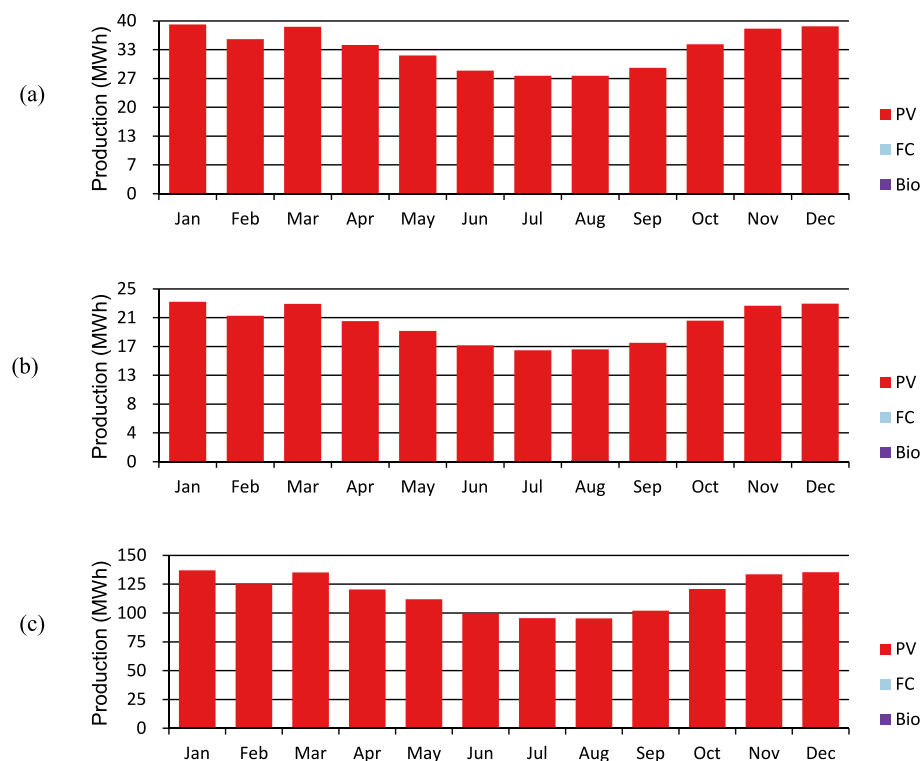


Fig. 11. Monthly average electricity production for optimal hybrid solutions for low-consumers (a), medium consumers (b) and high consumers (c) communities in scenario 2.

cell (2.29%). It is important to mention that the maximum electricity production occurred in the period December–January (dry season). It is also judicious to highlight that the minimum value was recorded in the

period July–August (rainy season). This observation can be explained by the fact that during the rainy season, the sky was cloudy. In addition, it was concluded that by increasing the load, the PV was more in demand

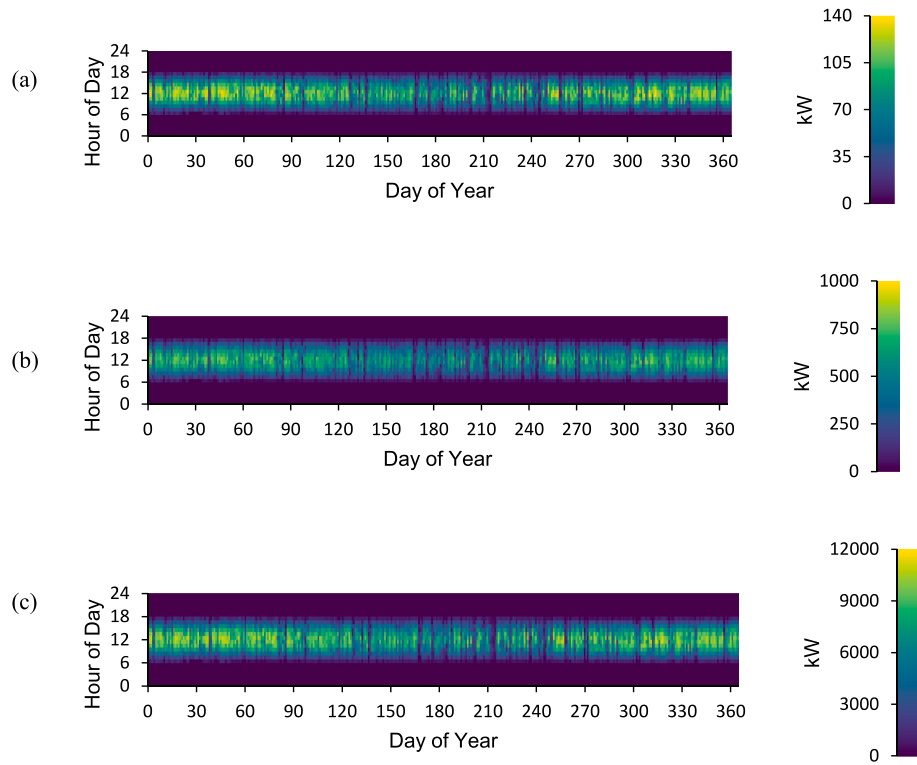


Fig. 12. PV power output for low-consumers (a), medium consumers (b) and high consumers (c) communities in scenario 1.

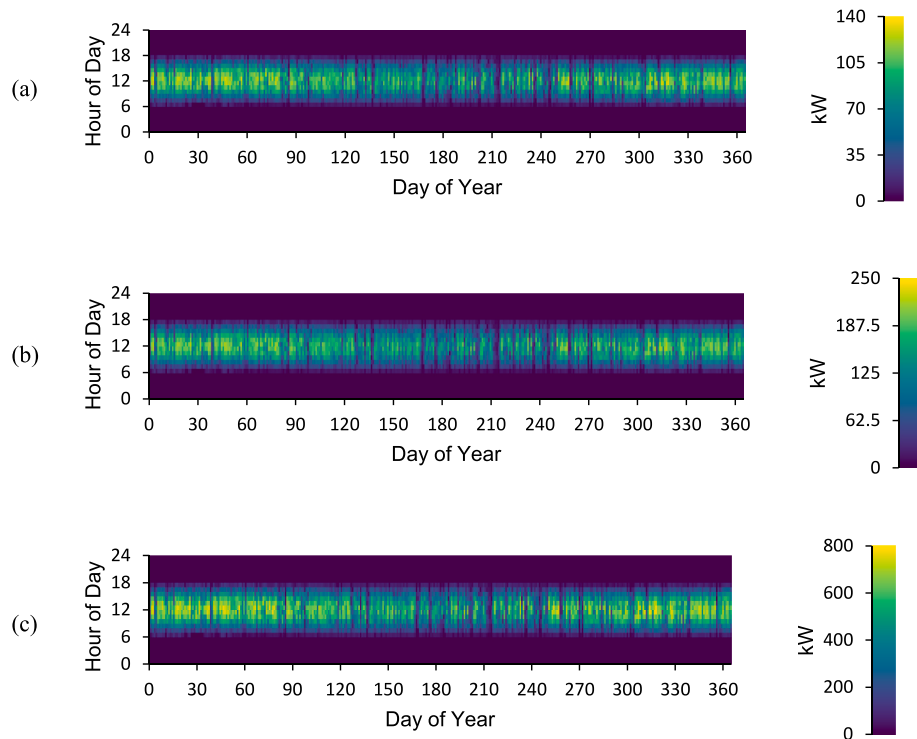


Fig. 13. PV power output for low-consumers (a), medium consumers (b) and high consumers (c) communities in scenario 2.

for the production of electrical energy than the biogas generator and the fuel cell.

Fig. 11 presents the monthly average electricity production for optimal hybrid solutions for each type of household electricity demand community in scenario 2. It was observed that the component with the highest annual electricity production was the PV array module

(99.97%), followed by the biogas generator (0.03%). The maximum and minimum electricity production also occurred in the periods of December–January and July–August respectively. From this scenario result, it was concluded that the storage system mainly consisted of battery bank since no power was produced by the fuel cell.

In Fig. 12, the PV power output for low-consumers (a), medium

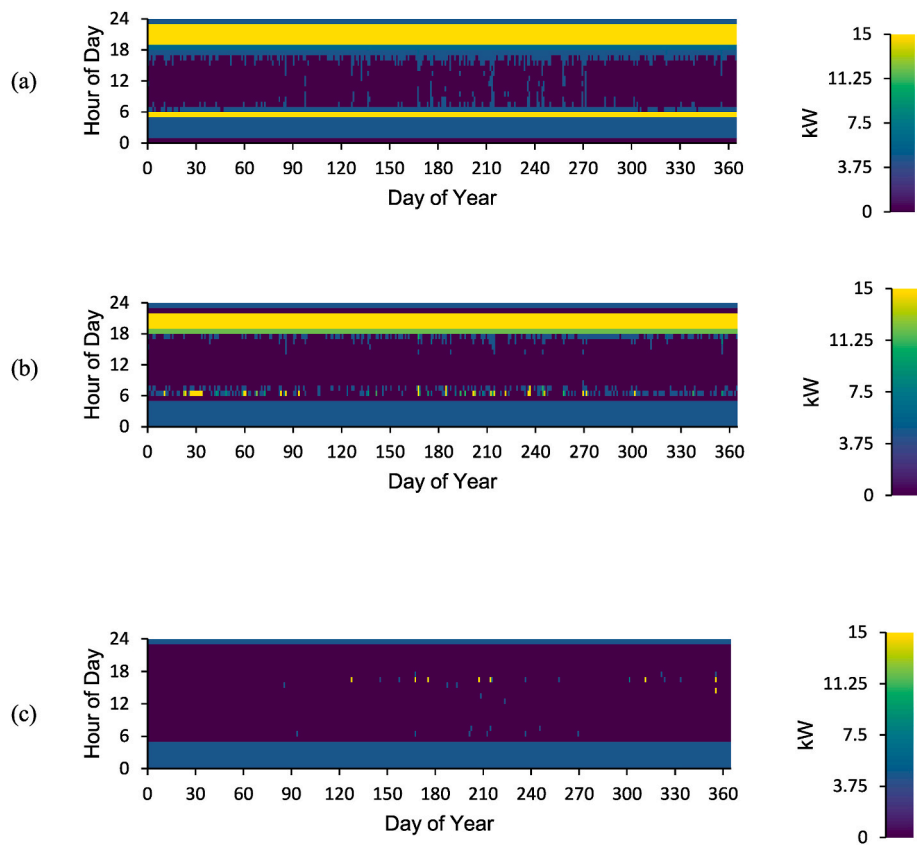


Fig. 14. Power output from biogas generator system for low-consumers (a), medium consumers (b) and high consumers (c) communities for scenario 1.

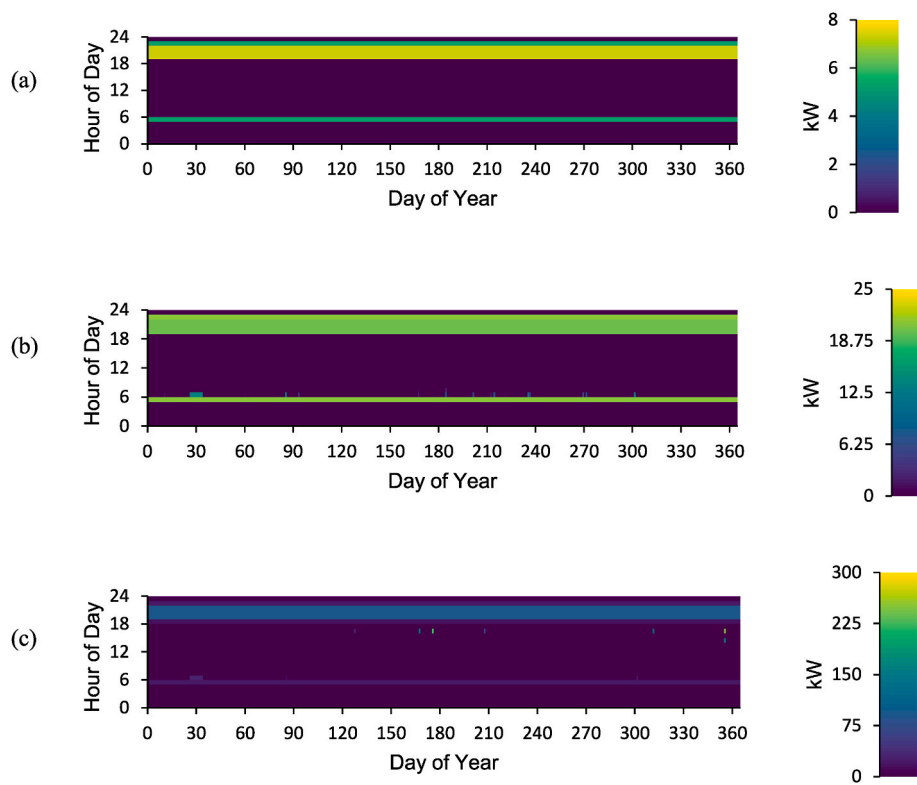


Fig. 15. Power output from fuel cell for low-consumers (a), medium consumers (b) and high consumers (c) communities for scenario 1.

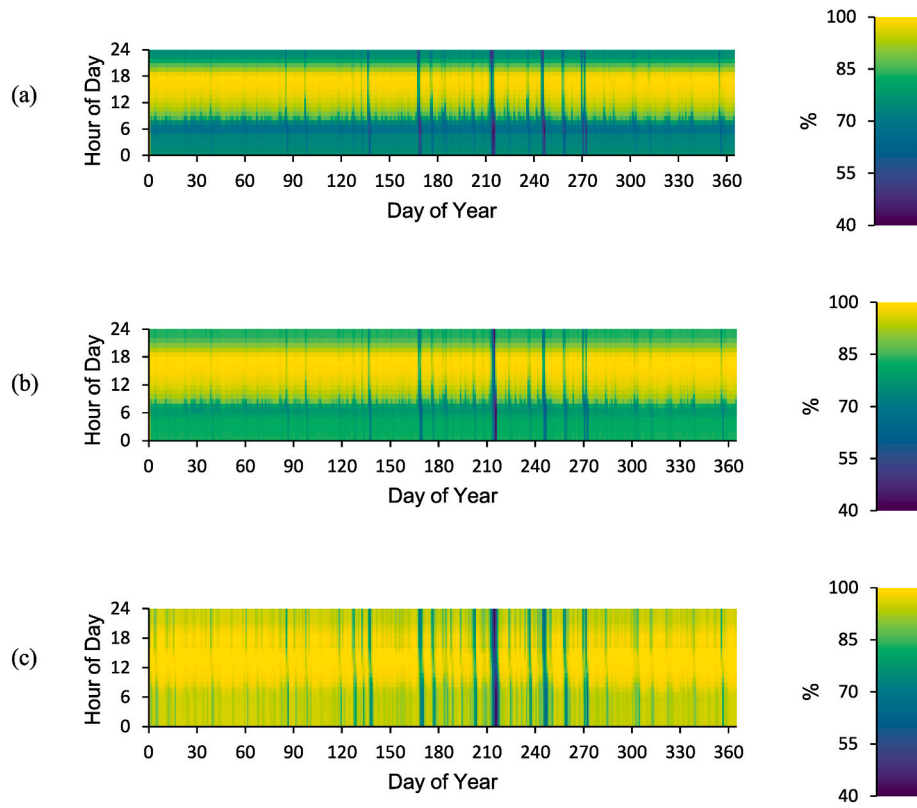


Fig. 16. State of charge of batteries throughout the year for low-consumers (a), medium consumers (b) and high consumers (c) communities for scenario 2.

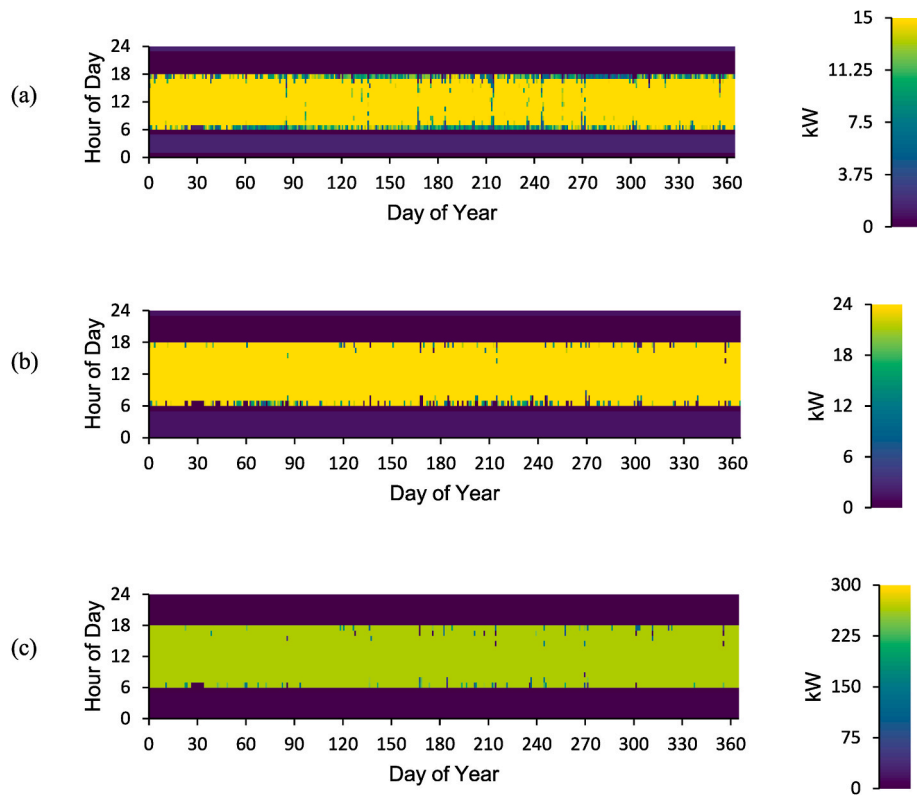


Fig. 17. Electrolyzer input power throughout the year for low-consumers (a), medium consumers (b) and high consumers (c) communities for scenario 1.

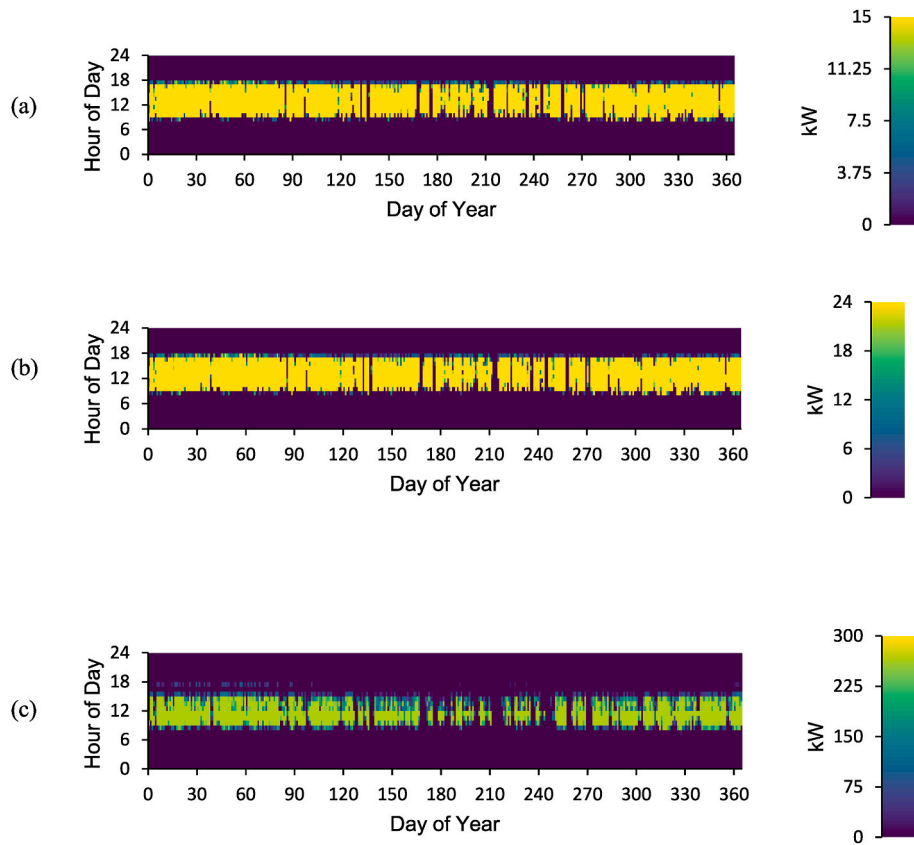


Fig. 18. Electrolyzer input power throughout the year for low-consumers (a), medium consumers (b) and high consumers (c) communities for scenario 2.

consumers (b) and high consumers (c) communities of scenario 1 was presented. An observation of this Figure showed that most of the electrical energy production from PV modules occurred between 9 a.m. and 4 p.m. throughout the year. It was also depicted a peak of the production at 12 p.m. for all the categories of consumers, which can be explained by the fact that the sun was at its zenith. It was also noticed that the power outputs from PV for high and medium consumers communities were respectively 85 and 12 times higher than the power output from PV for low consumers. These solutions also suggested that between 5 p.m. and 8 a.m., the PV modules were off due to the absence of sun.

In Fig. 13, the PV power output for low-consumers (a), medium consumers (b) and high consumers (c) communities respectively of scenario 2 was depicted. It can be also concluded that most of the electrical energy production from PV modules occurred between 9 a.m. and 4 p.m. It is also important to highlight from Figs. 12 and 13 that the PV power output was higher with scenario 1 when compared with scenario 2 for medium and high consumers communities. In terms of comparison, the power outputs from PV for high and medium consumers communities were respectively 5 and 3 times higher than the power output from PV for low consumers.

Fig. 14 presents the power output from biogas generator system for low-consumers (a), medium consumers (b) and high consumers (c) communities respectively for scenario 1. It was observed here that the maximum power output of the biogas generator took place between 7 p.m. and 11 p.m. throughout the year for low and medium communities consumers. A part of the electricity production from biogas generator occurred around 6 a.m. For the special case of high consumers community, the production occurred around 12 p.m. and 6 a.m. Since, the power production of the biogas generator was negligible for scenario 2, it was not presented in this paper. It was concluded from Fig. 14 that the biogas generator operated when the PV modules were off during certain periods of the day.

Fig. 15 depicts the power output from fuel cell for low-consumers (a),

medium consumers (b) and high consumers (c) communities respectively for scenario 1. Here, it can be noticed that the electricity production of fuel cell mainly took place between 7 p.m. and 11 p.m. for all the categories of communities throughout the year. The electricity production from fuel cell obtained by conversion of the chemical energy of Hydrogen also occurred when the PV modules were off because at these particular periods of the day, this electrical energy production was used for the operation of some equipment (TV, refrigerator, electric iron, lighting and security bulbs). It can be concluded that the fuel cell in the hours of the day when the amount of production capacity was reduced, helped the system to compensate for the deficit. In addition, the production was higher for high consumers when compared with low and medium consumers. Since from Table 12, it was noticed no power production from fuel cell, the scenario 2 was not presented in this work.

Fig. 16 shows the state of charge of batteries throughout the year for low-consumers (a), medium consumers (b) and high consumers (c) communities for scenario 2. It demonstrated that the batteries were more filled between 10 a.m. and 8 p.m. throughout the year for low and medium communities. This situation reflected the higher electricity demand of the different communities during that particular period of the day. It is also very important to highlight that the batteries were less charged between 9 p.m. and 9 a.m. because the PV modules were not in operation and it was the energy stored in the battery during the day which was used at this period.

Fig. 17 exhibits the electrolyzer input power throughout the year for low-consumers (a), medium consumers (b) and high consumers (c) communities for scenario 1. Here, it was observed that the maximum input power occurred between 6 a.m. and 6 p.m. throughout the year for all the consumers communities. Indeed, the electrolyzer produces Hydrogen through a chemical process (electrolysis) capable of separating the Hydrogen and oxygen molecules of which water is composed using electricity. The electrolyzer input power came from the electrical energy produced by the PV modules during the day. It was higher for the

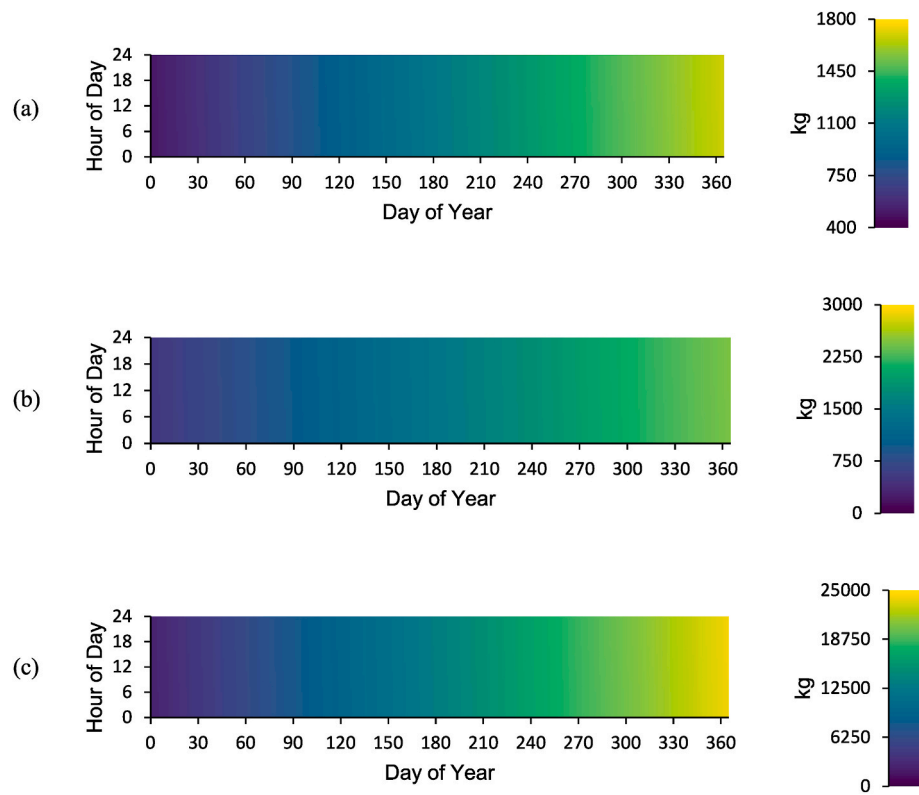


Fig. 19. Hydrogen tank level throughout the year for low-consumers (a), medium consumers (b) and high consumers (c) communities for scenario 1.

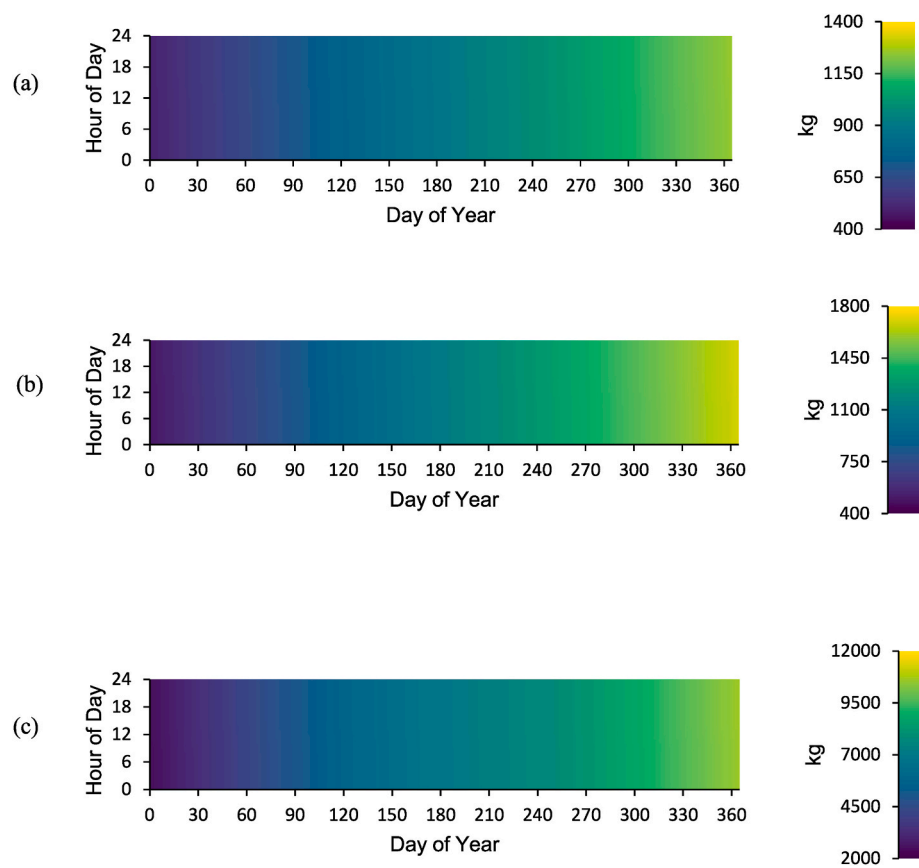


Fig. 20. Hydrogen tank level throughout the year for low-consumers (a), medium consumers (b) and high consumers (c) communities for scenario 2.

Table 13

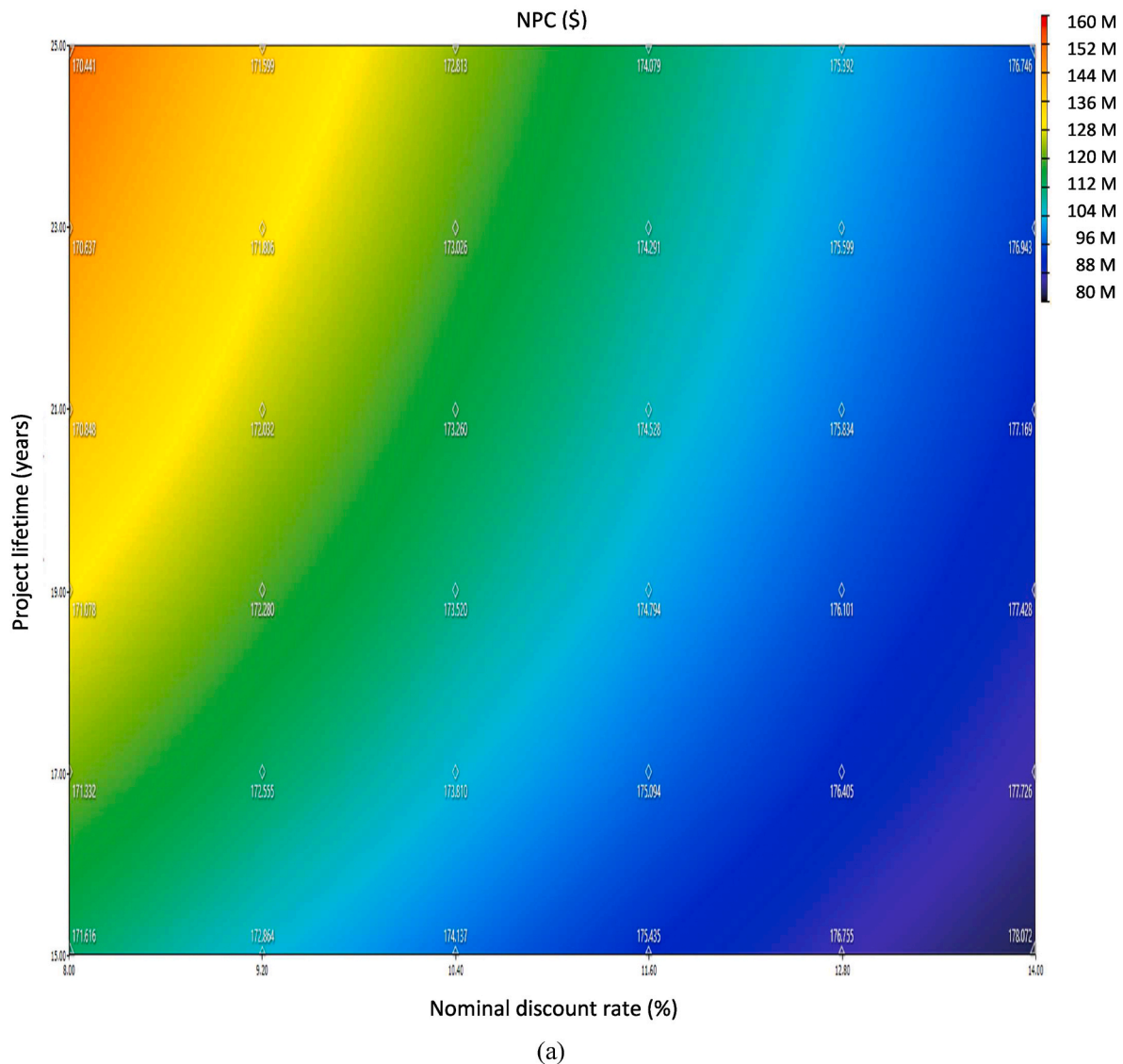
Comparison between scenario 1 and scenario 2 for carbon dioxide and carbon monoxide emissions for low, medium and high consumers communities respectively.

Community	Carbon Dioxide emission (kg/yr)			Carbon Monoxide emission (kg/yr)		
	Scenario 1	Scenario 2	Reduction (%)	Scenario 1	Scenario 2	Reduction (%)
Low-consumer	5190	1.250	99.98	4.240	0.01390	99.67
Medium-consumer	15,531	0.265	99.99	4.490	0.00294	99.93
High-consumer	49,617	0	100	6.05	0	100

Table 14

Cash-flow by component for low-consumers (a), medium consumers (b) and high consumers (c) communities for the scenario 1 and scenario 2.

Component	Scenario 1			Scenario 2		
	Low-	Medium-	High-	Low-	Medium-	High-
PV panel	\$100,917	\$615,300	\$8.35 M	\$96,860	\$164,080	\$573,876
Biogas generator	\$1.03 M	\$828,400	\$444,600	\$2200	\$600.00	\$0.0
Electrolyzer	\$3000	\$4800	\$53,200	\$3000	\$4800	\$53,200
Fuel cell	\$8.21 M	\$15.6 M	\$188 M	\$0.0	\$0.0	\$0.0
Hydrogen tank	\$750,000	\$750,000	\$3.75 M	\$750,000	\$750,000	\$3.75 M
Converter	\$1078	\$3124	\$55,860	\$2669	\$5666	\$39,738
Battery	–	–	–	\$22,320	\$47,570	\$442,634

**Fig. 21.** Surface plot: net present cost, superimposed: cost of energy for low (a), medium (b) and high consumers (c) communities (scenario 1).

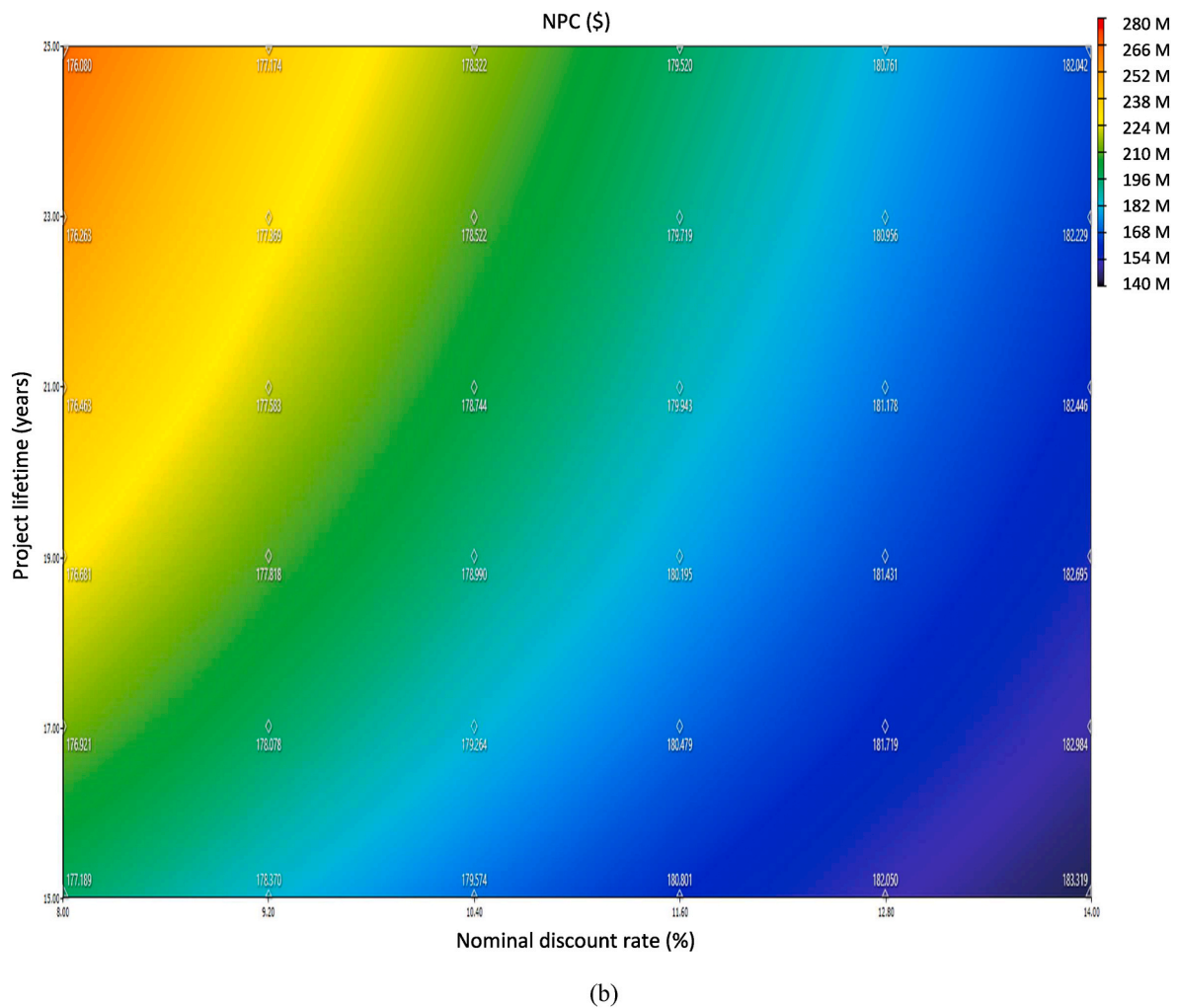


Fig. 21. (continued).

high consumers community when compared with the other communities.

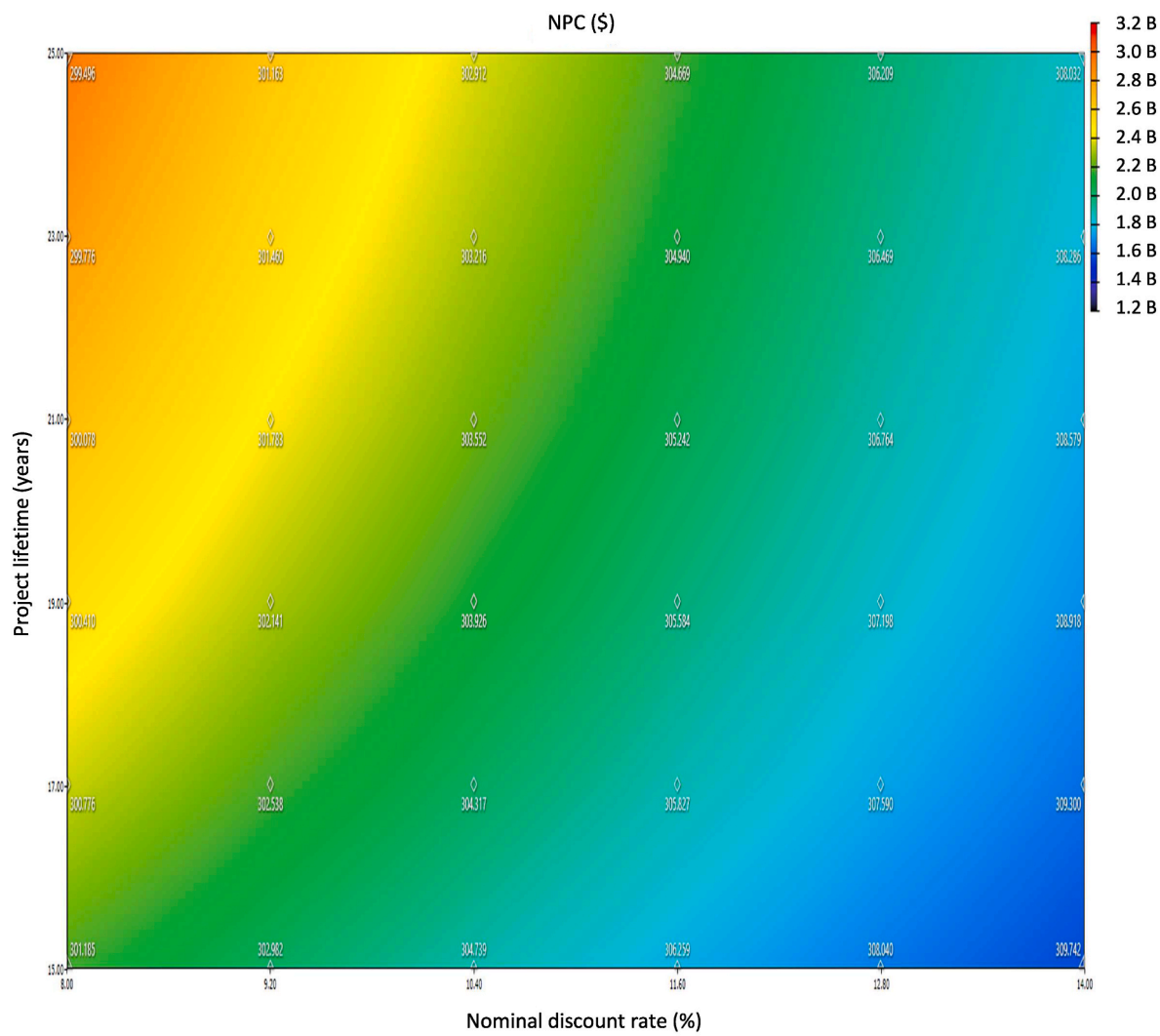
Fig. 18 presents the electrolyzer input power throughout the year for low-consumers (a), medium consumers (b) and high consumers (c) communities for scenario 2. The electrolyzer input power took place here between 9 a.m. and 3 p.m. for all the consumers communities. It can be therefore concluded that it was more pronounced for scenario 1 when compared with scenario 2.

Figs. 19 and 20 depict the level of Hydrogen produced by electrolysis in scenario 1 and scenario 2 respectively for low, medium and high consumers. As it can be seen, a part of the Hydrogen was used in fuel cell and the other part could be stored for other purposes such as the operation of the ongoing tramway lines project [79]. It is worth mentioning that this generation of Hydrogen was due to the high production capacity of solar panels at noon. An overall observation of Figs. 19 and 20 revealed that the Hydrogen was more produced in scenario 1 when compared with scenario 2. This was explained by the use of the combination of fuel cell, electrolyzer and Hydrogen tank in the architecture of scenario 1.

Table 13 presents the comparison between scenario 1 and scenario 2 for carbon dioxide and carbon monoxide emissions for low, medium and high consumers communities respectively. The results demonstrated that PV, biogas generators, fuel cells and electrolyzer can produce renewable and sustainable energy and tackle the emission problem. The values of carbon dioxide and carbon monoxide were substantially reduced with scenario 2 when compared with scenario 1.

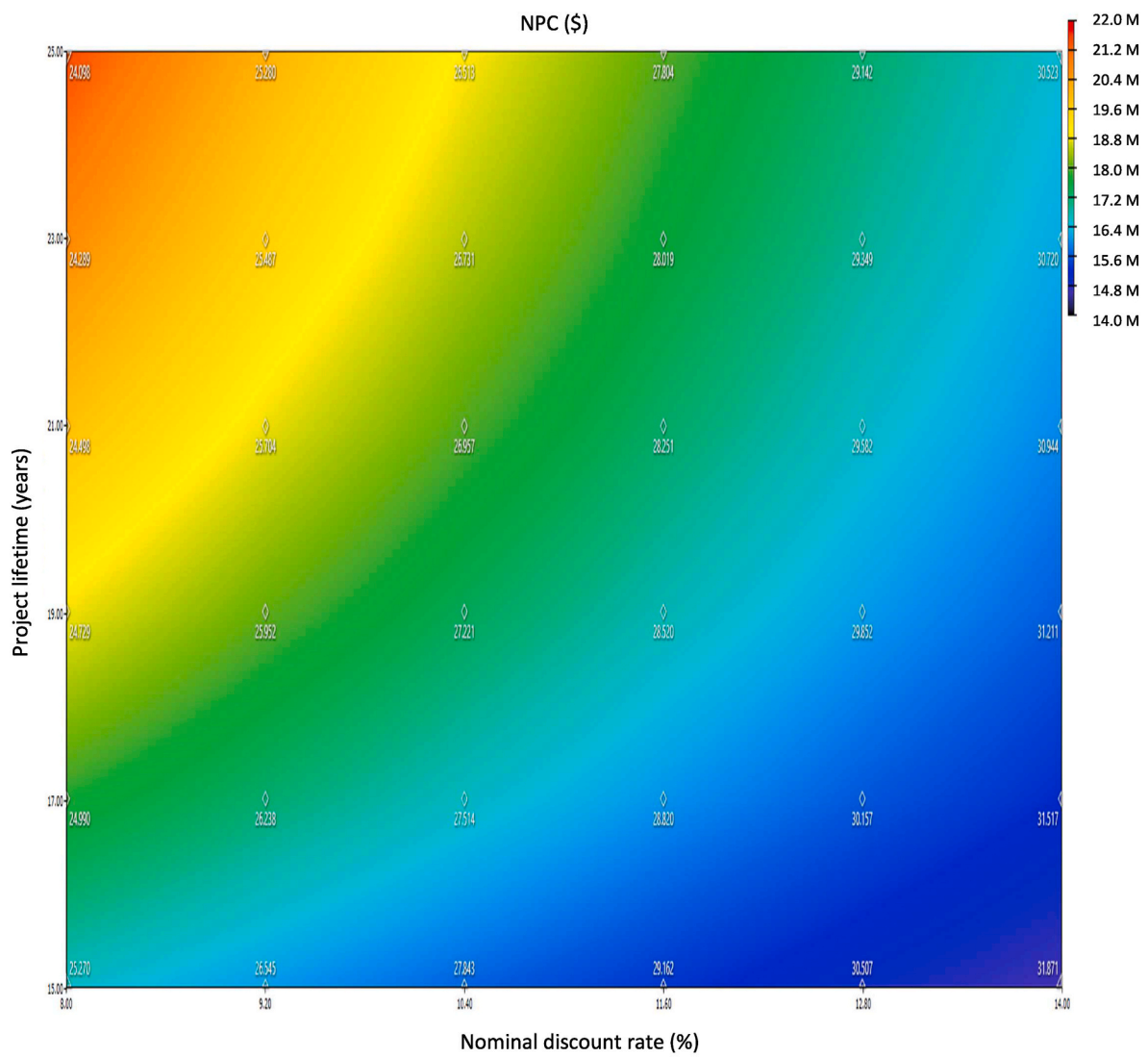
Table 14 presents the cash-flow by component for low-consumers (a), medium consumers (b) and high consumers (c) communities for the scenario 1 and scenario 2 respectively. It was observed in scenario 1 that the fuel cell represented the major contribution of the total cost throughout the life of the project followed by the biogas generator (for low and medium consumers) or Hydrogen tank (for high consumers), and the PV modules. Concerning the scenario 2, the Hydrogen tank was the component with the major contribution of the total cost followed by the PV module and the batteries.

Figs. 21 and 22 illustrate the impact of changes in the discount rate and the cost of energy. Indeed, the knowledge of these two parameters allows policy makers and investors to guide their decisions in choosing their investment policy in the field of renewable energy. In this study, the surface plots of net present cost, superimposed: cost of energy for low (a), medium (b) and high consumers (c) communities (scenario 1 and scenario 2 respectively) were examined. In these aforementioned figures, the area represents the net present cost when the cost of energy is superimposed on the area. It can be concluded that when the discount rate increased, the cost of energy also increased for all the categories of consumers of scenario 1 and scenario 2. It was also observed that increasing the discount rate and the price of fuel implied the increasing of the cost of energy. In order to attract investment in renewable energy, Cameroon's government could apply lower fuel prices and apply a 0% interest rate to capital invested to make electricity available and accessible to the rural population.



(c)

Fig. 21. (continued).



(a)

Fig. 22. Surface plot: net present cost, superimposed: cost of energy for low (a), medium (b) and high consumers (c) communities (scenario 2).

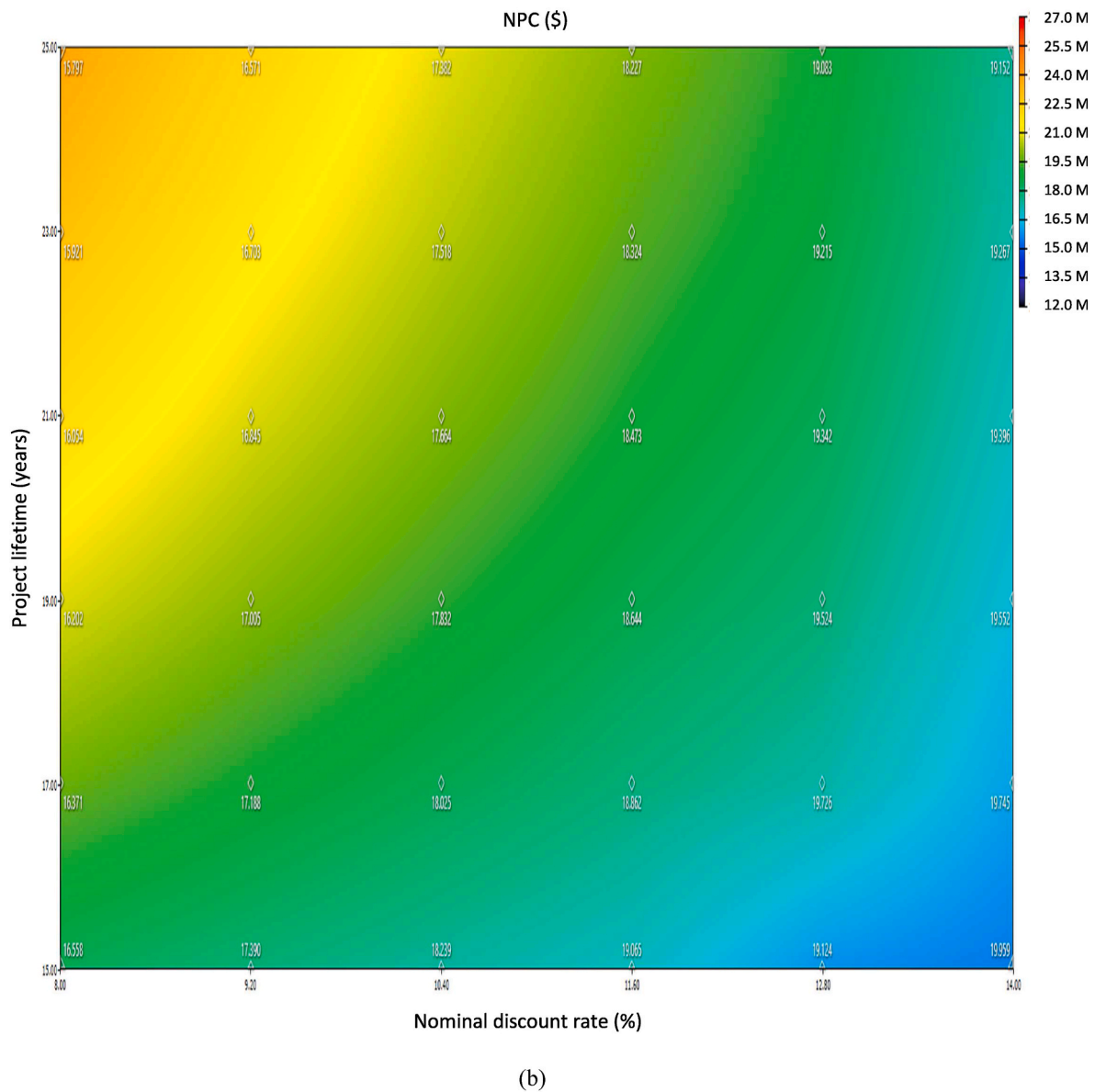


Fig. 22. (continued).

4. Conclusion

Electricity production and Hydrogen generation by the use of PV module, biogas generator, fuel cell, and electrolyzer were carried out in the city of Maroua in the Far North region of Cameroon in this work. Two scenarios of hybrid energy systems were considered, namely PV/Fuel Cell/Electrolyzer/Biogas (scenario 1), and PV/Battery/Fuel Cell/Electrolyzer/Biogas (scenario 2). As storage systems, two configurations were taken into account; a battery bank and the association of electrolyzer, fuel cell and Hydrogen tank. It was found that:

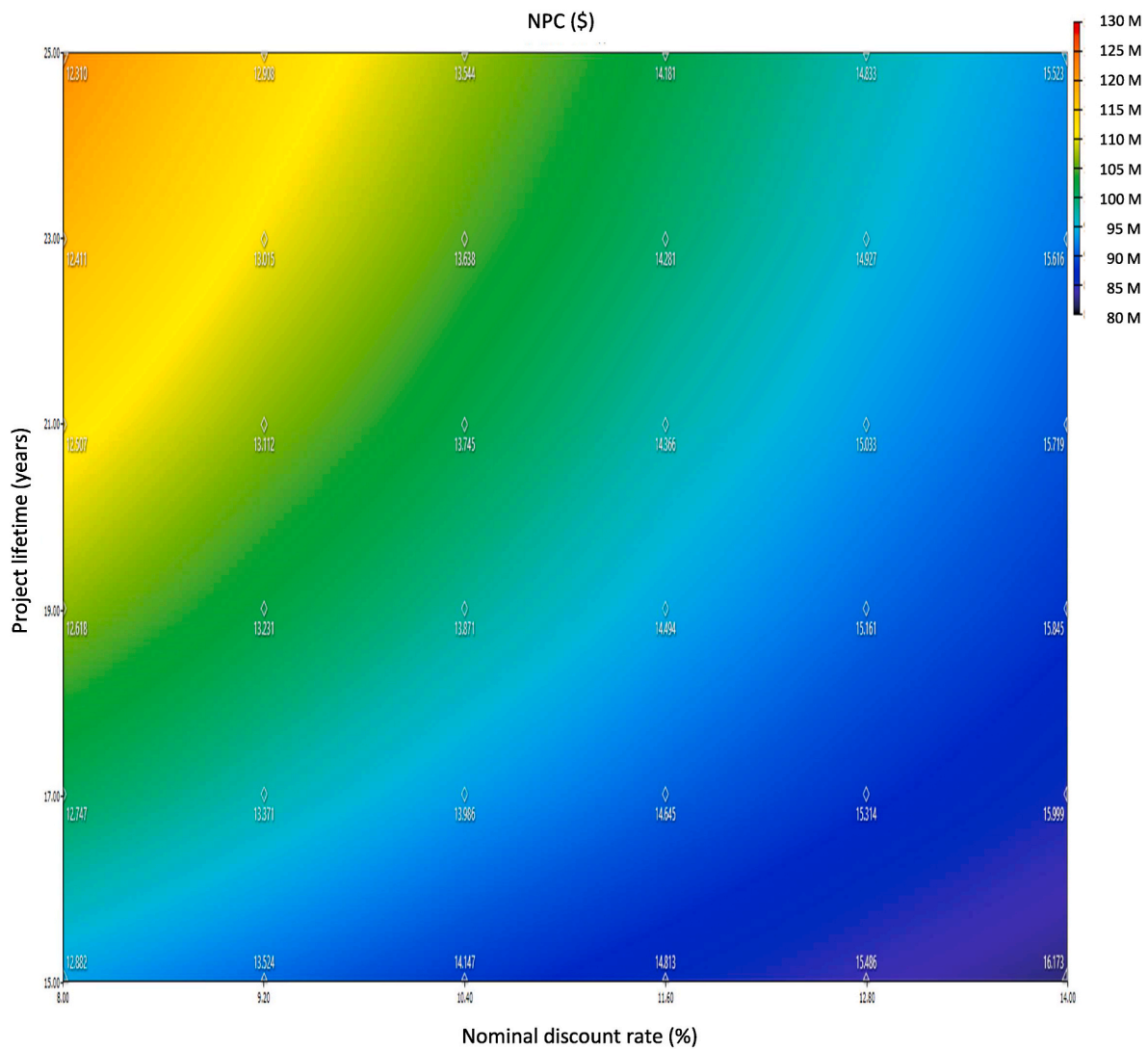
- The use of a battery bank in the architecture led to the minimum cost of energy for each electricity demand community;
- The increasing of the discount rate and the price of fuel implied the increasing of the LCOE;
- To attract investment in renewable energy, Cameroon's government could apply lower fuel prices and apply a 0% interest rate to capital invested to make electricity available and accessible to the rural population;

- The overall emissions of carbon dioxide and carbon monoxide were relatively low for scenario 2 when compared with scenario 1;
- The Hydrogen was more produced in scenario 1 when compared with scenario 2;
- The Hydrogen produced in this work could be beneficial first of all for use in fuel cell and secondly for the operation of the ongoing tramway lines project in Cameroon [79].

Incorporating hydroelectric and wind turbines in the present scenarios will constitute future works. In addition, performing a comparison of simulation with and without tilt angle and tracking system was left as a future investigation.

Credit author statement

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(c)

Fig. 22. (continued).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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